

Germanic, Slavic and Baltic ‘thousand’ once more

Rémy Viredaz, Geneva
20.10.2025

1. Introduction

1.1. Problem statement

The word for ‘thousand’ in Baltic, Slavic and Germanic languages is a well-known puzzle, with forms that are obviously related but annoyingly divergent (Bopp 1853: 446; Derksen 2015: 473 f., 2008: 503, Comrie 1992: 792-795, Jakob 2024: 740-745, Kroonen 2013: 554, Ross–Berns 1992: 621): Lithuanian *tūkstantis*, Latvian *tūkstuots*; Finnish (loan) *tuhat*; Old Prussian acc. pl. *tūsintons*; Slavic **tysęti*, **tysęti*; Gothic *þūsundi*, Old Norse *þúsund* without umlaut but also *þúsand*, *þúsind*, *þúshund*, *þús-hundrað*, Old English *þūsend*, Old High German *dūsunt*.

The most commonly posited PIE preform is now perhaps **tuH-s-(o)nt-* > **tūs(o)nt-*, but the word underwent a number of irregular changes, more or less well explained, on its way to the daughter languages, especially in Baltic, and several points remain disputed¹.

2. Germanic

See forms in Ross–Berns 1992: 621, 682, Lühr 1998: 890 f., Gorbachov 2006: 2, Braune 2023/1886: 229 §274.

2.1. Vocalism

Ross–Berns 1992: 621 posit no less than five protoforms : (1) **þūsundjō-* f., nom. sg. **þūsundī*, (2) **þūsunda-* n., (3) **þūsanda-* n., (4) **þūsandja-* n., (5) **þūsinda-* n. However, this is a minority view. Only the first of these types is sure, while the semblance of others can often be explained by secondary innovations, i. e. post-Proto-Germanic accidents, see van Helten 1906: 122 f., Gorbachov 2006: 2, 13.

¹ It’s a pleasure to thank Anthony Jakob, Juho Pystynen and Douglas G. Kilday for their crucial help with various points of this paper (fn. 2, 32, 54, 76, 78).

2.1.1. Gothic

Goth *þusundi* subst. f., nom. pl. *-jos* (Streitberg 1920: 136 f.), reflects PGmc **þūsundī*.

2.1.2. Old Norse

2.1.2.1. Nominative ending

ON *þúsund* f.², without umlaut, is not phonetically regular (if from **-ī*)³. Even morphologically, **þūsundī* contrasts with other **-ī/*-jō*-stems like **wulgtī* ‘she-wolf’, **armī* ‘sleeve’ → **-īz* > *ylgr*, *ermr*.

a) Van Helten, l. c. 122 explains it by an irregular shortening of final **-ī*⁴ supposed to have occurred early enough for the resulting **-i* to undergo the primary apocope, the one that predates umlaut. However, that’s not the likeliest chronology, because that would imply that Gothic had escaped this irregular shortening and only suffered the later, regular shortening of final long vowels. Now, the archaism of Gothic is not generally due to its having split off earlier (in fact, earlier emigration is more of a factor of innovations), but to a larger extent merely to its being attested earlier.

b) One alternative explanation would be to favour the opposite chronology: 1° Regular pan-Germanic shortening of final long vowels⁵. 2° “Post-Gothic”, irregular loss of **-i* in this word, both in North and West Germanic (below), early enough to predate umlaut, hence ON *þúsund*.

Numerals are relatively prone to irregular phonetic reductions. However, *þúsund* is neither particularly frequent, nor particularly long. Another factor of these reductions is that numerals often appear in contexts in which a numeral is expected so that it potentially contrasts mostly with other numerals. Still, ‘thousand’ does not really fit this criterion, because of its frequent use for an indefinite large quantity (2.4).

c) *þúsund* is both nominative and accusative, unlike the regular type, Gothic nom. *mawi* : acc. *mauja* ‘girl’ (~ Lith. *saldī*, *sáldžiq* ‘sweet (fem.)’, OCS *bogynji*, *bogynjō* ‘goddess’). This suggests that the word shifted to the **i*-inflection for some

² Thanks to Douglas G. Kilday for his help in understanding the intricate diachrony of **-ī-*, **-ō-* and **-ī/jō-* stems in ON.

³ Gorbachov 2006: 2 suggests that nom. sg. *þúsund*, gen. sg. *þúsundar* are phonetically regular, which is mistaken on both accounts as far as we have understood ON phonology.

⁴ Van Helten does not specify whether this shortening would be regular or irregular, but we have found no rule to that effect anywhere in Noreen’s grammars.

⁵ This shortening is pan-Germanic, but not Proto-Germanic, cf. **-ō* > Goth *-a* vs. OHG, ON *(*)-u*.

reason, i. e. the reflex of **þūsundī* (with its paradigm) gave way to what would have been the reflex of a virtual **þūsundiz*. Most old **i*-stems do not show *-r* in the nominative singular in ON, and many do not show umlaut, perhaps because PGmc **-in* was treated as **-i* i. e. the final **n* did not “protect” **i*, **u* from early shortening (whereas it did protect **ō*, i. e. **-ōn* was not treated like **-ō*). Indeed, van Helten 1906: 123 speaks on an “Übertritt des *ī*-stammes in die fem. *i*-Deklination”. The reason for this change of inflection is probably that the model of the feminines in nom. sg. **-ō* : nom. pl. **-ōz* induced the feminines in nom. sg. *-ī* to take the nom. pl. **-īz* of the **-i*-stems (< **-ijiz* < PIE **-eyes*).

2.1.2.2. Inflection

The inflection is not phonetically regular either⁶, but it’s consistent with the hypothesis of a shift to the **-i*-stem inflection:

a) Nom. pl. *þúsyndir*, *þúsynder*, with *i*-umlaut.

b) Gen. sg. *þúsundar*, dat. pl. *þúsundum* with no trace of the **j* of PGmc **þūsundjōz*. Regular sound change would have yielded umlaut throughout, but perhaps the stem of unumlauted acc. sg. *-und* (if regular, cf. 2.1.2.1) spread to the nominative and the rest of the inflection.

c) Occasional analogical extension of *u* in West Norse pl. *þúsunder*, of **y* in the sg. *-ind* forms below (van Helten, l. c.).

2.1.2.3. Variations in vocalism

An unattested **þúsynd* (2.1.2.2 c) seems necessary to account for the forms in *-ind* – unless one is wishing to admit a PGmc **þūsenda-* (as if from PIE **tūsento-*) in addition to **þūsundī* (< **tūsntī*).

Although van Helten (l. c.)⁷ sees the *i* in Runic Swedish *þusind*, Old Danish *t(h)usin(d)* as a spelling for /y/, this type of spelling seems too frequent for that explanation and reflects perhaps a linguistic reality, namely a dissimilation of unaccented **y* to *i* after *ú* in this word. The dissimilation may have taken place either in the singular **þúsynd*, analogical of pl. *þúsynder*, or directly in pl. *þúsynder* > **þúsinder* before the creation of *þúsind*.

Rather than analogy, an alternative explanation for the **-ynd* (> *-ind*) forms (including Old Frisian below) might be that PGmc **þūsundī* regionally and optionally escaped the irregular shortening of **-ī* mentioned above. Yet such an

⁶ Gorbachov’s square brackets and asterisks in 2006: 2 do not mark unattested forms, but forms that have drifted away from the *devī* type (or *mawī* type in Gothic).

⁷ With references to Noreen, whose only evidence for Old Norse is a hapax *ind* for *ynd*. [We have not been able to see van Helten’s *Altnostfrk. Gr.*]

early areal feature straddling the main dialectal division between North and West Germanic may not be likely.

As for the *a* of OSw *þúsand*, ODan *tusand*, it results from an “obvious” influence of *hundrað* ‘hundred’ according to van Helten (l. c.).

The other spellings OSw *þúsæn*, ODan *thusænd* are not clear to us.

There is some extent of unpredictable variation *i* ~ *a* ~ *u* (a forerunner to the reduction to *ə*) in Old Norse and Old Swedish (Noreen 1904: 164-167).

2.1.3. High German

Explanations similar to those cited for Old Norse above account for OHG *thūsunt*, *dūsunt* and Old North-East Franconian *thūsint* (van Helten, l. c.). The loss of the ending (whatever the reason, see 2.1.2.1 for Old Norse) in **þūsundī* → OHG *dūsunt* is probably responsible for the latter becoming indeclinable or shifting to other inflection types (cf. Gorbachov 2006: 2). Influence of *hunt* ‘hundred’ caused an occasional shift to the neuter gender (van Helten, l. c. 123; Lühr 1998: 891).

The rare MHG variant *tūsant* (Ross–Berns l. c., quoting Mausser 1933: 525; not in Lexer), besides *tūsent*, *tūsunt*, cannot be explained in the same way as suggested for ON *þúsand* by van Helten (above, 2.1.2, end), since ‘hundred’ does not show forms in *a* in German. Given its date and the lack of instances such as **thūsant*, **dūsant* in this word in German (Ross–Berns 1992, 620), we must rather have to do with an inverse spelling of ⁺*tūsənt*, cf. *ābant* ~ *ābent* [ə] ‘evening’ (cf. Mausser 1933, 525).

The neuter gender of *tūsent* in Notker and the indeclinable use of *thūsunt* by Otfrid are analogical to *hunt* ‘hundred’ (van Helten, l. c. 123).

2.1.4. English

OE *þūsend* cannot reflect Comrie’s (1992: 621) PGmc **þūsenda-* (which would be **þūsinda-* and cause umlaut). The change of inflection suggested above 2.1.2.1 c would let us expect umlaut (cf. **anstiz* ‘favour’ > OE *æst*, *ēst*, Ol Frisian *enst* vs. ON *ást*, *óst*, OS OHG *anst*). Van Helten 1906: 122 seems to imply that *e* of *þūsend* represents the regular umlaut of **u*. It seems to us that OE *e* in this position could spell *ə* and reflect any PGmc short vowel, cf. Brunner 1942: 123.

-and in the modern spelling *thousand* is not old but appears in Middle English – not in the literature according to *MED* (except for one line in Chaucer⁸),

⁸ *Canterbury Tales* A 1954 (Wright 1956). – We don’t know what *MED* means by “thousand [thousand]”. The text has *thousand* in all four modern editions we have looked up, with no mention of a *varia lectio*, and so does the Hegwrt manuscript photographed in Ruggiers 1979: 100.

but abundantly in other sources (among many other spellings: *eMED* s. v. *thousand*).

2.1.5. Old Frisian

thûsend, with *e* to be explained as in Old English (2.1.4).

2.1.6. Methodological note

Linguistic reconstruction mostly strives to monogenesis, i. e. tracing cognate forms of the daughter languages to a single prototype⁹. In a privileged instance such as PIE **sweks* ‘six’, monogenesis is a welcome *conclusion* (namely, all the alleged evidence for initial variants **se-*, **we-*, **ks(w)e-* can be plausibly dismissed as resulting from post-PIE innovations on criteria *other* than the mere wish for monogenesis, Viredaz 1997). More often, however, monogenesis is only a working *hypothesis*, based on the general principle of parsimony (Occam’s razor), even if in many cases, like PGmc **pūsundī*, unicity of origin can only be achieved through a number of additional hypotheses, e. g. irregular sound changes, analogical levelings or borrowings. (There’s a whole typology of irregular changes, with some of them less irregular than others.) Nevertheless, we hold that the example of **sweks* confirms the general validity of the monogenetic principle.

2.2. Inlaut consonantism

2.2.1. The **-s-h-* tale

Some Germanic forms have often been pressed into service in favour of an etymology starting from PGmc **pūshundī*, PIE **tūs-k̑mt-ī* ‘Krafthundert’: Cleasby-Vigfusson 751 s. v. *pūsund*, Bugge 1888: 327, Streitberg 1896: 145; for further references see Bugge l. c., Pijnenburg 1989: 100, Lühr 1998: 892, Gorbachov 2006: 8. Despite all refutations, this view is still adhered to, with slight reservation, by Kroonen 2013: 554 s. v. **pūsundī*-. However, PIE **-sk̑-* would yield Gmc **-sk-* (because **tus*, **pūs* is not an independent word). An attempt to fix this problem is Szemerényi’s (1990: 240 f., 1996: 227) **tūso-k̑mtī* > **pūsa-hundī* (followed by Rix 1991: 225), which, however, only makes things worse even for Germanic in terms of irregular sound changes.

These forms are better explained otherwise:

ON *pūshund*, a rare variant of *pūsund*, is certainly due to popular etymology after **hund* ‘cent’ (Hirt 1896: 345, van Helten 1906: 122, Gorbachov 2006: 13). Another variant is *pūshundrað*. A problem for this view is that *pūshund* only turns

⁹ This may not belong to the usual or registered acceptations of the word, but it concords with its overall meaning and we’ve found it useful for the present discussion.

up in written documents after *hund* ‘hundred’ has gone out of use: *púshund* is not found in the earliest documents while *hund* (Cleasby-Vigfusson 292) is only found as first member of poetic compounds with a value that could be weakly translated as ‘very’. A solution is perhaps that *púshund* was initially branded as incorrect speech and only later found its way in the formal language of a few writers. That *púshund* might be older than *pūsund* is disproved by its non-occurrence in the earliest texts and especially by Gothic *pūsundi*.

The archaic form *thuschunde* that Jakob Grimm (1853: 385), followed by Kluge (who finds that the evidence “besonders schwer wiegt”, 1901: 491) and others (including Kroonen 2013, 554: with a tad of caution), wanted to read in the – extremely corrupt – glosses to the *Lex Salica*, is unfounded speculation: the spellings *-tius chunde*, *-tus chunde*, *-thus chunde*, *-tos chunde* of cod. 8 correspond to *-nu sunde*, *-t usunde*, *tho sunde*, *-no sunde* of the Heroldiana; another line of the same list shows in these two manuscripts *-thocundi*, *to condi* respectively; spellings in *ch* must be analogical of the word *chunna* (< **hund-n-*) which occurs several times in that list (van Helten 1900: 512, 514, 516, followed by e. g. Ross–Berns 1992: 621, Lühr 1993: 120 f., 1998: 894)¹⁰.

It’s remarkable how successful this **-s-h-*tale has been (e. g. Brugmann 1909: 48 f., Mausser 1933: 873) and how late it’s remained attractive to some scholars even though it has already been disproved in 1896-1900.

2.2.2. Verner’s Law

The voiceless **s* of PGmc **pūsundī* proves that the PIE accent was on the first syllable (unless perhaps one starts from PIE **tūtsn̥tī*, which we don’t, 5.1.2, or from **tūs-ḱm̥tī*, which is impossible, 2.2.1, 5.1.1).

Mausser 1933: 873⁽²⁾ insists that the voiced **d* of the third syllable proves that the PIE accent was final. This, however, is a misinterpretation of Verner’s Law. Verner’s voicing is regular after a non-initial syllable whatever the accent

While the initial accent of Lithuanian *tūkstantis*, Slavic **tysę/otji* is not significant because it might also be due to Hirt’s Law, the former initial accent implied by Gmc **pūsundī* must be inherited.

2.2.3. The *d*-less forms

In light of Goth *sunjis* ‘true’, *sunja* ‘truth’, OE *synne* ‘sin’ (vs. OS *sundia*, OHG *suntea* ‘id.’), Goth *sunjon sik* ‘to excuse oneself’, ON *synja* ‘to deny (a charge)’, *syn* ‘denial’ it seems that PGmc **-ndj-* is regularly reduced to **-nj-*, hence

¹⁰ Gorbachov 2006: 11, 13 cites the Salian forms as if they were *sprachwirklich*, which they are not, and analogical of **hund* ‘hundred’, which is not accurate (the source of the influence is *chunna*, a derivative of **hund*, see above).

PIE **h₁s-nt-ih₂*, **-yéh₂-s* > post-PIE **sntī*, **sntyās* > PGmc **sundī*, **sunjōz*, with leveling in different directions, and **snt-yó-* (Skt *satyá-* ‘true’) > PGmc **sunja-* (e. g. van Helten 1906: 123, Kroonen 2013: 492)¹¹.

In ‘thousand’, however, it seems that the re-generalization of **-nd-* is old and pan- or Proto-Germanic and that the final *-n* of some languages (Modern Swedish and Norwegian, sporadically in Old Swedish and Danish, in Middle English and in modern Frisian, Dutch and Low German dialects, Ross–Berns 1992: 621) is a later irregular reduction of **-nd*, because *d*-less forms are not found in the oldest documents (Gothic [not significant for paucity of examples], Runic ON; OHG, OS, OE).

2.3. Later accidents

We’ll skip a few curiosities that are too late to be relevant for the Indo-European etymology, e. g.:

Initial devoicing in MHG *tūsent*, ModG *tausend* vs. OHG *dūsunt*, see Bugge 1888: 325 f., Braune 2023: 228 §167 Anm. 8.

Irregular (?) reduction in Middle Dutch *duust*, *duyst* (beside *dusen(t)*, *dusant*, *dusentich*), Flemish *duust*, Ross–Berns 1992: 621, Lühr 1998: 891.

General or sporadic reduction **-nd* > *-n* (2.2.3).

Velar for dental in the MHG variants *tūsinc*, *tūseng*, *tūsig*, Ross–Berns 1992: 682 n. 324; compare perhaps *nd* > *ng* in Swiss-German, Zurich *tuusig*, *Aabig*, *Hund*, *hinder*, Bern *tuusig*, *Aabe* (general), *Hung*, *hinger* (countryside)/*Hund*, *hinder* (city)¹² ‘thousand, evening, dog. behind’ (Weber–Bächtold 1983, von Greyerz–Bietenhard 1984, s. vv.); the wider extension with ‘thousand’ perhaps by dissimilation¹³.

2.4. Frequency, meaning

Cleasby-Vigfússon 751 s. v. *púsund* note the following:

There is little doubt that with the ancient heathen Scandinavians (and perhaps all Teutons), before their contact with the civilised southern people, the notion of numbers was limited, and that their thousand was not a definite number, but a vague term, denoting a swarm, crowd, host (cp. the Gr. *μυρίοι*): in ancient lays it occurs thrice (Hkv., Em., Fas. i. 502), but indefinitely (...) [I]n the ancient popular literature, uninfluenced by southern writers, ‘púsund,’ as a definite number, occurs, we think, not half-a-dozen times. As the

¹¹ It could be asked whether Skt *satyá-*, Gmc **sunja-* is not old but back-formed on the feminine oblique stem. However, this is unlikely especially for the Skt word.

¹² Restored after other dialects that had remained closer to Standard German.

¹³ There is even an isolated spelling *zowsing* in Middle English (*eMED* s. v. *thousand*). Unlike the German instances, it is perhaps linked to the shift *-and* → *-ing* of the participle suffix, itself due to a confusion with the action noun suffix *-ing*.

multiple of ten duodecimal hundreds, ere the decimal hundred was adopted, ‘þúsund’ would mean twelve decimal hundreds (...) [I]n ecclesiastical writers, and in annals influenced by the Latin and the like, it [‘þúsund’] is frequent enough (...).

As regards the meaning, however, it is possible, and even likely, that the “vague” meaning is secondary rather than primary. This happens for other numerals as well, often in a hyperbolic sense, e. g. French *être à deux doigts de...* (very close), *haut comme trois pommes* (as tall as three apples, said of a small child), *Il fait toujours trente-six choses à la fois* (many; some say *trente-six mille*), *Je te l’ai déjà dit cent fois, C’est mille fois meilleur*.

2.5. Conclusion (Germanic)

Although not everything is clear, it seems reasonably possible that all Germanic forms go back to a single prototype **þūsundī*.

3. Slavic

3.1. Final

OCS *tysęšti* and *tysqšti* < **-tji* f.¹⁴, **jā*-stem, (Comrie 1992: 793).

Eastern and Southern languages have forms like R *týsjača*, SCr *tìsuća*, as if from Slavic **-tja*, by analogical leveling.

Western languages and most of Slovenian show masculine forms in *-c*, Sln *-č*, as if from **-tjb*. See Comrie 1992: 794 f. for the areal extension and the actual forms. As explained by Vondrák 1906: 337 f., 1924: 424, Vaillant 1958: 648, this **-tjb* has arisen from the genitive plural **-tjb* of the **jā*-stem, which was originally used for all cases of multiples of 1000 from 5000 onwards, such as **pętъ* (‘5’, inflected) + **tysętjb*/**tysqtb* (gen. pl.).

3.2. Vocalism

3.2.1. Distribution

The vowel of the second syllable goes back to either **ę* or **ǫ*. Apart from those languages or dialects that have replaced ‘1000’ by loans or paraphrases, **ę* is reflected in East Slavic, Polish and older Slovak, **ǫ* in Serbo-Croatian and Slovenian; both are found Old Church Slavonic (below) and both have somehow merged in Czech: see Comrie 1992: 793, with more detail in 793-795.

¹⁴ The nominative singular in *-šta* given in some grammars is not found in OCS itself, as far as Diels 1963: 218, Anm. 8 is aware of. This certainly holds as well for the examples of *-šta* in Savvina kniga and Psalt. sin. adduced by Pijnenburg 1989: 100.

Old Church Slavonic, as said, has both *tysęšt-* and *tysqšt-*, often in the same manuscript: see Diels 1932: 93 for their distribution. According to Rosenkranz 1955: 107, it seems that *tysqšti* (more common) is Macedonian and *tysęšti* Bulgarian; it's not clear to us if this conclusion is arrived at based on the manuscript evidence or on extrapolation from the distribution in the modern languages. Although Modern Bulgarian and Macedonian use *(x)iljada* (3.4), note that an obsolete Bulgarian *tísešta* is quoted by Vasmer s. v. *ты́сяча*, after Mladenov¹⁵.

3.2.2. Chronology

Applying the principle of parsimony (2.1.6) 1° within Slavic and 2° between IE dialects, one expects 1° that Proto-Slavic had only one of the two protoforms **tysętji* and **tysq̃tji*, and 2° that this protoform was **tysętji* < **-intī* with zero grade in accordance with Germanic **pūsundī* (in spite of East Baltic *tūkstantis*, *tūkstuoš*)¹⁶.

However, unless we have missed something, none of the two expectations above is obvious from Slavic-internal evidence, such as the distribution of the two spellings in OCS manuscripts, the areas of the two kinds of reflexes in the daughter languages, or the outcome of other instances of competition between suffixal **ę* and **q̃*.

Thus, OCS *byšęšte-je/byšq̃šte-je* 'τὰ μέλλοντα, the future', OCz *pro-byšúcný* 'useful' (compared by Gorbachov 2006: 11) corresponds to Lith *būsiqs* (and *-antis*), fem. *-anti* adj. 'future', which means that **byšq̃tj-* is inherited and **byšętj-* a preliterate innovation (perhaps due to nom. sing. m. **byšę* < Pre-Slavic **būšjans*: while the OCS alternation *-ę* : *-q̃št-* is well preserved in participles of **-je*-presents, **byšq̃št-* had lost its verbal basis and thus its participial status so that its isolation may have made it more vulnerable). As for OCS *goręšt-/gorq̃št-* 'burning' (compared by Comrie 1992: 792), *gorq̃št-* (S-Cr *gòrūćī* 'hot, burning') is the older form while *goręšt-* is built on the (new) present indicative stem *gori-* (according to e. g. Meillet 1924: 196¹⁷).

¹⁵ I don't know if this *tísešta* could be a mere coinage by Mladenov or his predecessor Gerov, based on OCS *tysęšta*. But perhaps this would rather have produced **tísašta*, based on *tysq̃šta*.

¹⁶ Some scholars indeed admit that only **ę* is old in Slavic in this word, but for invalid reasons (PIE **tūs-k̑mtī*, Comrie 1992: 792) or undisclosed reasons (Gorbachov 2006: 11, 13).

¹⁷ Cf. Meillet–Vaillant 1965: 235. – Le russe moderne a à la fois *gorjáčij* 'chaud, brûlant' et *gorjúčij* 'combustible, inflammable' (Ščerba–Matusevič, *Dict. r.-fr.*).

Although **tyšetji*/**tysotji* is not a participle, it has been suggested that **ę* → **o* in it was due to the influence of the (present active) participles¹⁸. There are two suffixes for these participles, **-ot-* (with present indicatives in *-e-*, **-je-* and even postvocalic *-i-*) and **-et-* (with present indicatives in postconsonantal **-i-*) (Diels 1963: 232 f., cf. 244-281)¹⁹. Thus it can be assumed that *-ot-* was felt as the normal or general suffix and *-et-* as linked to clear specific conditions, which may have lead speakers to think that **tyšetji* was a mistake for ***tysotji* and favour the latter by hypercorrection.

Still, it is a bit frustrating to have no Slavic-internal factual evidence for this conclusion, which is nothing but our a priori hypothesis (merely based on the principle of parsimony, see beginning of paragraph), and, worse, not even to have a more obvious explanation than the above for the assumed innovation.

3.3. Consonantism

As is known, the reconstruction **tūs(o)nt-* (1), suggested by Germanic **pūsundī* (2), and morphologically more likely than others (3.3.1), does not directly account for the Slavic *-s-*, because it should yield Sl. **tūšant-/tūšint-* > **tyxot-/tyšet-*. Many proposals have been attempted to solve this difficulty²⁰.

3.3.1. Misguided attempts

Hirt 1896 and others (including Vaillant 1958: 647, Lühr 1998: 898) think that the Slavic word is borrowed from Germanic. But that's digging a hole to fill another, because it's now the Slavic *-t-* (and *-ę-*) that would be problematic.

It has also been thought that the Germanic, Slavic and Baltic word was a loan from an unknown language (*Wanderwort*, see 5.3), which may help if the borrowing postdates the RUKI law. While the assumption of a German loan can

¹⁸ Vondrák (quoted by Lühr 1993: 126²⁸, Gorbachov 2006: 11) thinks that – Comrie 1992: 792 starts from **ę* (cf. n. 16) and suggests a reinterpretation as a participle as the possible cause for a hesitation between **ę* and **o*; *ibid.* 793, however, he suggests reinterpretations of BSl **-mt-* (our **-nt-*), first as a suffix subject to ablaut, and subsequently as the suffix of the present participle. (Cf. Endzelīns 1923: 365 f., 1971 : 183, as a means of explaining the alleged change **-amt-* → **-ant-* in Baltic.) – Gorbachov 2006: 13 starts from **tyšetji* as well and adds: “Slavic **tyšetji* is secondary and relatively rare (participle-influenced?); no *o*-grade would have been originally possible”; cf. *ibid.* 11: “purely morphological assimilation of **tyšetji* to the fem. participial paradigm?”.

¹⁹ We omit the few instances of former athematic presents as well as *vědě* ‘know’ (see Diels 1933: 233, 262, 278 f.). – The rule survives in the daughter languages (Serbo-Croatian: Vaillant GSC 188 f.; Russian: Garde 1980: 317). The distribution is similar in Baltic (Lithuanian) for present active participles in *-ant-* and *-int-* (Kurschat 1876).

²⁰ The following is just a sample. See more detail and more criticism in Pijnenburg 1989: 100-104, Lühr 1998: 892-901, Gorbachov 2006: 7-9.

easily be refuted because the Proto-Germanic form is known well enough, that of a Wanderwort is hard to falsify, not because it is better founded but, on the contrary, because we know nothing of the source language.

Comrie 792 starts from “a form like **tū(s)-k̑mt-*”, which “will give regularly **tys̑et-*”, thus assuming an **s*-less prototype for Slavic. However, **-sk̑-* would not account for the Germanic reflex (2.2). This means that the Germanic (*-s-*), Slavic (**-k̑-*) and Baltic (**-sk̑-*) words would go back to three different forms in (late) PIE.

Pijnenburg 1989: 104 suggests **tūts̑ntī*, which would indeed account for the Slavic and Germanic reflexes (cf. Gmc **wīsaz* ‘wise’ < **weissa-*)²¹. But it’s morphologically problematic, and does not really account for Baltic *-st-* (see 5.1.2).

We long thought that the solution was a PIE **tūst(o)nt-*, with dissimilatory loss of the medial **t* in all languages except East Baltic. The RUKI law does not apply in Slavic before a stop (Meillet–Vaillant 1934: 32, 34, Vaillant 1950: 29); or more exactly, it did apply, but a later innovation changed Slavic **š* of any origin to *s* (or *z*) before a stop (cf. Viredaz 1997: 115, with reference to Pedersen)²²; later, **š* became **x* in Slavic before a vowel or semi-vowel (and still later, **x* became *x’* > *š* before front vowel and *j*). Thus, it sufficed to assume that the dissimilation in question predated the change of Satem **št* to Slavic *st* and that would have accounted for the Slavic *s*. But this **tūst(o)nt-*, too, is morphologically problematic, because we’ve never found a satisfying explanation for the internal **-(s)t-* (5.1.3).

3.3.2. Influence of ‘hundred’

Finally, given the weakness of the proposals above (3.3.1), “[i]t seems that we cannot avoid assuming that the word for thousand was influenced by [...] ‘hundred’” (to use the words of Derksen 2015: 474 s. v. *tūkstantis*, in contrast to 2008: 503 s. v. **tys̑ti*: “The exact formation is unclear”), before the latter was replaced by the Iranian loan **sata* > **soto* > **s̑to*²³. This solution has also been that of Aitzetmüller 1978: 140, Lühr 1993: 123, bottom, 124, bottom; Gorbachov 2006: 10).

²¹ As if from **weid̑s̑to-*, but more probably a later v̑ddhi-derivative of **wissa-* ‘known’ < **wid̑s̑to-*.

²² In our opinion, the circumstance “of any origin” constitutes the “proof” that Vaillant 1950: 29 deemed was lacking.

²³ Our explanation of *s̑to* in 2020, 416 f. is utterly unlikely, and it is better to follow Šaxmatov’s view (quoted by Vasmer) of a reduction of **soto* < **sata*. However, there remain 1° the contradiction between the reduction in **soto* and the lack of reduction in **des̑et̑* (except later in bound phrases for teens or tens), despite the fact that the latter is longer and more frequent, and 2° the contradiction between Romanian *sută*, which seems to suggest a borrowing at a Slavic stage close to **suta*, and the Slavic reduction, which seems to have occurred at a later stage **soto* > *s̑to*. We hope to return to this problem elsewhere.

Rather than **tyšę̃tji*, **tyxę̃tji* → **tysę̃tji*, **tysę̃tji* after **sę̃to* (thus Gorbachov 2006: 10, and, presumably, implicitly the other authors cited as well), or **tūxintī* → **tūsintī* after **sinta* ‘hundred’, we think it more likely that the change took place earlier as **tūšintī* > **tūčintī* after **činta*²⁴.

3.4. Loans

Various Slavic languages or dialects – including, alas, Bulgarian and Macedonian – have replaced the old word by loans from neighbouring languages (Greek, German, Hungarian, Romance) or paraphrases (Comrie 1992: 793-795).

4. East Baltic

Lithuanian *tūkstantis*, with many variants, Latvian *tūkstuotis*, with variants²⁵, see Kurschat 1876: 263, 269, Bezzenberger 1877: 183, 333, Endzelīns 1971: 183, Fraenkel 1965: 1135, Pijnenburg 1989: 99 f., Comrie 1992: 792 f., Smoczyński 2018: 1532. Ordinal Lith *tūkstantas* and others, Latv *tūkstošais*, Kurschat 265, Endzelīns 1971: 185.

The connection with the Germanic and Slavic forms raises many questions²⁶.

4.1. -k-

4.1.1. Epenthesis

East Baltic -k- in this word must be epenthetic (Stang 1966: 109, Derksen 2018: 474).

²⁴ Although it has long been customary to posit **š* for the Balto-Slavic stage of PIE **k̑* > Baltic **š*, Slavic *s*, this is not the most plausible reconstruction. For reasons of diachronic phonetic likelihood independent of the word for ‘thousand’, we posit PIE **k̑*, **k̑s*, **k̑t*, **k̑st* > Satem I **č*, **čš*, **čt*, **čšt* > Satem II **č*, **tš*, **čt*, **št* (Viredaz 1997: 114¹⁰⁻¹¹) > Balto-Slavic **č*, **čš*, **čt* (or **št*), **št*. For PIE **k̑* at least, this is now indeed commonly done. – In this paper, we have followed the majority and write **k̑*, **k* for our usual **k₁*, **k₂*. This does not entail a change of opinion, however (“**k̑*” was not a palatal in PIE but it became one in Satem dialects including Luwian). – As for the RUKI law, for obvious phonetic reasons, the path must have been **s* > **š* > Indo-Aryan *ś* rather than **s* > **š* > other languages *š* (see also 4.2 for Baltic). Note, however, that Stang’s **š* is just a conventional symbol for the altered **s* without reference to its exact phonetic value (cf. Stang 1966: 16).

²⁵ As is known, the official spelling uses *o* while comparativists mostly write *uo* after Mühlénbach–Endzelīns (Petit 2020: 56, 58), which is closer to the pronunciation and emphasizes the symmetry with *ie*.

²⁶ Our approach of taking the problems one by one may not be the best way to reach a solution after all. We keep it as a mode of exposition, however.

An epenthetic *k* or *g* is frequent in (East and West) Baltic before a cluster of sibilant + dental stop, and even found sometimes before a sibilant generally (Stang l. c. 108-110). In *tūkstantis*, the velar not only may, but must be secondary because, if it were inherited (as Smoczyński suggests, see below 5.4.3), the expected sibilant after it would be *š by RUKI law. Indeed, a *k*-less variant is attested once each in Old Lithuanian (gen. *tūstancios*, Postilė, 1599) and Old Latvian (*tuustofsche*, *ALEW* s. v. *tūkstantis*). More precisely, Daukša's Postille has *-kst-* generally (e. g. 125₃₀, 297₁₇) and *-st-* only once (123₃₁), which is “vielleicht nicht Druckfehler” (Senn 1966: 213).

4.1.2. Root *tūk-*?

Since the velar epenthesis is not regular but sporadic, one might think that an additional factor such as an influence of the Baltic root **tuk-* ou **tūk-* ‘grow fat’²⁷ (thus Comrie 1992: 792) would be welcome. However, no such factors seem to have taken place in other examples of epenthesis (though we have not made a systematic search), and we would not like to make *tūkstantis* an exception.

In any case, this root *tuk-* cannot be the *proper* etymology (as Smoczyński suggests, see below 5.4.3), were it only because of the following *s* (4.1.1).

4.2. *-s-*

The sibilant in the Lithuanian word for ‘thousand’ is *s*, not *š*, at all times and in all dialects, with the following exceptions as far as I know, all in the old language:

– In Bretkūnas’ translation of the Bible (started 1579, published 1590), Bezzenberger 1877: 81 observes that the word often has the spelling *tukſchtantis*, in which *ſch* is sometimes corrected to *f*, sometimes not. This means that *-kst-* was the

²⁷ There are three different views: (a) root *tuk-/taūk-*, acute secondary, *LIV*² s. v. **teuk-* and Smoczyński 2018: 1533 s. vv. *tūkstantis*, *tūkti*; (b) root **tūk-/tauk-* according to Endzelīns 1923: 584³, Gorbachov 2006: 12, Derksen 2015: 474 s. v. *taukaĩ*, less convincing; (c) Adolphi 1685: 204 “still” distinguished Latv *tūkstu*, *tūkt* ‘swell’ and *tūku*, *tukt* ‘grow fat’ (Endzelīns, l. c.), which implies two original Baltic roots **tūk-* and **tuk-*, the latter with a nasal-infix present. – This root (or one of these roots) is found in Baltic (Derksen 2015: 459 f., 474), Slavic (Derksen 2008: 499 f.) and Germanic (**peuha-* ‘thigh’). The root **tū-* < **tuH-* of other languages is also found in Slavic (Derksen 2008: 503 f. s. v. *tyti* ‘grow fat’, Vasmer–Trubačev s. v. *отавы* ‘aftermath’) and even Baltic (Lith *tūlas* ‘many’, Smoczyński 2018: 1534). – Perhaps an original **tuk-* acquired an optional variant **tūk-* in Baltic with expressive lengthening triggered by the influence of **tū-*, and early Latvian secondarily introduced a semantic distinction between *tuk-* and *tūk-*; the phonetically regular merging of present **tūnk-* > *tūk-* with non-present **tūk-* > *tūk-* may have been the reason that prompted the addition of the present suffix *-st-* in the latter root. – In Indo-European, a root **teuHk-* (Derksen, Kroonen) would be anomalous; as for **teuk-*, it might be a (dialectal) expressive variant of **teuH-* or just a different root.

correct form, but the higher frequency of the sequence *kšt* in the language led to Bretkūnas's hesitation, as Bezenberger notes.

– *tukſchtanczias* (acc. pl.) is found in the Wolfenbüttel Postille (1573; *ALEW*).

– *tūkſteriopq* ‘thousandfold’ is found in Daukša's Postille (1599; *ALEW*; see 4.1.1 for other spellings there)²⁸.

The old Finnic *š seen in e. g. Finnish *tuhat*, Mari *təžem* is not, in our opinion, a sufficient proof for a shibilant *š in the *Baltic* source language(s) (see 7.3 below).

The preservation of PIE *s after *i*, *u* is regular in Baltic, in which the RUKI law does not apply fully, although the conditions for the reflexes *s* or *š* are unclear, see Stang 1966: 13-16, 94-99.

This *s* is not a Baltic innovation (such as PIE *s > Balto-Slavic *š or *š > Baltic *s* with Brugmann I 781 or Hamp 1973: 173), but an archaism, and Stang l. c. 15 cites a similar retention in Nuristani²⁹, pointed out by his Oslo colleague Morgenstierne in 1926: 56 and later in 1975: 340.

In the case of Nuristani, the exceptions are probably borrowings from Gāndhārī (Middle Indic of the Dardic type) (Morgenstierne 1975: 340). In the case of Baltic, the reasons for the double treatment are unclear (Stang 1966: 96-99)³⁰.

We do not exclude that *s had an altered pronunciation after *i, *u in Early Baltic, but our claim is at least that these allophones had not become phonemic *š as they had in Slavic, Iranian and Indic.

In spite of many shared innovations, some of them remarkable, notably in the domain of prosody, Baltic and Slavic had remained different dialects inside the

²⁸ There are examples of a spelling *s* for anteconsonantal *š* in older texts (Stang 1966 : 99 f.). However, this cannot apply to ‘thousand’ in Bretkūnas’ text, both because the spellings and occasional correctopns show the opposite situation and because the modern language has *s*, not *š*.

²⁹ Mistakenly referred to there as a part of Dardic.

³⁰ Hamp 1973: 175 f. holds that the reflex of PIE *s after *i, *u in Baltic is *s* before a back vowel (including *a*) and *š before a front vowel (in fact, the suffixes *-ỹs/-is* < **-ijas* or *-ẽ/-ė* < **-ijā*). He then lists the “clearest attested forms”, that is, those that conform to his law. However, even allowing for analogical leveling (in unpredictable direction) between a derivative and its basis (even if not attested in Baltic), there remain the exceptions *vėtušas* ‘old’ and *aušrà* ‘dawn’, *aũšta*, *aũšti* ‘dawns, to dawn’ (< **aus-ro-*, **au(s)-skė-*, **aus-ti-*) on the one hand, *ausis* ‘ear’ on the other (whose dual *ausi* is even inherited, PIE **ausī* < **h₂us-s-ih₁*, though the singular replaces a PIE neuter **ausos* < **h₂eus-os* preserved in Slavic, Irslinger 2008: 339). Furthermore, all examples of the suffixes **-ija-* or **-ijā-* are Baltic only or Balto-Slavic only and thus likely to postdate RUKI. – It has not been possible in preparing this paper to gather and discuss all the literature devoted to the Baltic reflexes of *s after *i*, *u*.

Balto-Slavic language community, as results from some early differences such as perhaps 3 sg. present indicative **-et* vs. **-eti*, 1 pl. **-ome* vs. **-omos*; perhaps Lith *vanduõ* vs. Sl *voda* ‘water’ as if from **wondô*, **wod^hô* for **wodô(n)*.

4.3. -t-

On the question of East Baltic *-st-* vs. *-s-* of other languages, see 5 below.

4.4. Vocalism

As we have seen, Germanic reflects a (late) PIE **tūs-ŋt-ī* and Slavic continues either **tūs-ŋt-ī* as well or (less likely) **tūs-ont-ī* (3.2); West Baltic doesn’t testify (as OPr *tūsintons* may be borrowed, 6.4).

For (East) Baltic **-ant-i-* (stem class uncertain, see 4.6), various origins can be thought of, mainly these two:

a) PIE **tūs-ont-*, of which **tūsŋtī* was the feminine. A stem in **-omt-* or **-ont-* is almost universally posited. Enlargement of consonant stems through **-i-* is well known in European languages³¹, especially Balto-Slavic (after acc. sg. **-m̃ > *-in*). PIE **dek-m̃t-* ‘ten’ > Lith *dešimt(i)-* has partly remained a consonant stem (Endzelīns 1971, 182), but ‘thousand’ matches this, too (in dialects: Zinkevičius 1981: 64.).

b) PIE **tūs-ŋt-ī*, with zero grade of the first suffix as in Germanic, then a reshaping of both suffixes, yielding **tūs-ant-is*. This is (part of) Gorbachov’s solution, which we shall present and discuss later (5.4.5-5.5.0).

4.5. The nasal

Although many authors posit **-mt-* for Proto-Baltic – because of Old Prussian *tūsintons* but also *etymologiae causa* because of PIE **(d)k̑mtó-* or Proto-Baltic **čimta-* ‘hundred’ – the valid evidence points to **-nt-* only (6.2).

4.6. The final³²

Lith *tūkstantis* was a feminine in the old language (Bretkūnas, 1579-1590: genitive *-ties*) and it has been a masculine since the old language (Daukša, 1599) until today (genitive *-czio*, *-čio*). Some dialects have feminines in *-čią* or *-tė*.

The shift to masculine may be due to the analogy of *šimtas* ‘hundred’.

For the rest, it’s difficult to determine the original inflection from Lithuanian evidence alone. (See, however, Gorbachov 2006: 6.)

³¹ Starting from dat. sg. **-ei* (Meillet, Meier-Brügger).

³² Thanks to Anthony Jakob for his help in relation with the variations of the final within Lithuanian, as well as various other topics in this paper.

Comparison with Slavic and Germanic suggests the two possible PIE origins seen above (4.4).

5. East Baltic *-st-*

The hardest nut to crack is the internal cluster *-st-* of East Baltic as opposed to the *s* of Germanic and Slavic (and **š* in Fennic). Many explanations have been attempted³³.

5.1. “Preformist” attempts

There have been attempts to find a solution in Proto-Indo-European itself.

5.1.1. **-s-k-*

A common old view was that a compound stem **tūs-komt-* could account for the reflexes of all three branches Germanic, Slavic and Baltic. In reality, it accounts for none of the three (or just for East Baltic if this had earlier had **tūstintī*). See in particular the bibliography and criticism by Pijnenburg 1989: 100-103, Lühr 1998: 882-894, Gorbachov 2006: 8; see also 2.2.1 above.

5.1.2. **-ts-*

Pijnenburg 1989: 104 f. suggests a cluster **ts*, thus **tūtsntī* and **tūtsontī*.³⁴

This would work fine for Germanic and Slavic, but the metathesis required for East Baltic is unsupported (only **ks > sk* is attested; one might adduce Lith *stirna* ‘roe deer’ or *stumbras* ‘wisent’ but there is no indication that they go back to **ts-*).

5.1.3. Our first attempt: **-st-*

For a long time, we thought we had found the perfect solution (and this is the very reason we finally decided to write an article on this word): PIE **tūst(o)nt-*, with dissimilation of the medial **t* in all languages except East Baltic.

In Slavic, the dissimilation would have postdated the Slavic regular change of **št* to *st* (3.3.1)

For a short time, we thought the word could be a compound **tū(s)-st-(o)nt- < *tuh_x(s)-sth₂-(o)nt-* from the root of Latin *stare*, but this hardly makes sense.

³³ Gorbachov’s category of “amusing” theories (2006: 9) has been dispatched in our paper under various headings, no doubt for lack of humour on our part.

³⁴ Brief criticism by Gorbachov 2006: 9, Jakob 2024: 141.

5.2. Other “ingenuous” attempts

Another option is to start from PIE **tuHs-(o)nt-* > **tūs(o)nt-*, which accounts for Germanic and (indirectly, 3.3.2) Slavic, and try to find a plausible reason why PIE **-s-* would have become **-st-* in East Baltic.

5.2.1. *tūkstas*

A few authors have derived their solutions from the ordinal *tūkstas** (attested is the definite *tūkstàsis*), supposed to have been the model for a reshaping of the cardinal.

Gauthiot 1906: 3 f. thinks that East Baltic, at one point, had both the cardinal **tūšamtis*, from **tūs-komt-* (5.1.1, 4.5), and an ordinal **tūkstas*, formed directly on the root **tūs-*.

Būga 1908: 138 derives *tūkstas* from **tūskstós* < **tūsiko-stó-s* (cf. MHG dial. *tūsig*, Gk *χίλιοστός*).

Lühr 1993: 128, 1998: 901 thinks that **tūstas* is a shortening of **tūsantas* on the model of **šimtas* ‘hundredth’.

We have little doubt, however, that *tūkstas**, *tūkstàs-is* is merely a shortening of *tūkstantas*, *tūkstantàs-is*³⁵ and thus does not explain the *-st-*.

5.2.2. *-sta-*

Several attempts involve the inchoative verbal suffix *-sta-*: see 5.4 below.

5.3. “Defeatist” attempt

I mean the approach that gives up searching for explanations within IE and assumes a *Wanderwort* of unknown origin. This includes Pedersen 1943: 185 , Stang 1966: 282, 1972: 59, Jakob 2024: 144 f.

For sure it’s also a positive thing to try and gather potential evidence for regional substrates

Still, this approach has the downside of explaining little and not yielding an etymology, because the phonology and lexicon of the assumed source language are unknown. It’s only plausible if no Indo-European interpretation is.

5.4. Inchoative *-sta-*

A number of proposals involve the Baltic inchoative suffix *-sta-*, supposed to continue PIE **-sk^e/o-*. (This origin of the Baltic verbal suffix is advocated by Leumann 1942, Gorbachov 2006: 11, 2014, and accepted with reservation by

³⁵ Similar judgment by Kalnins 2015: 36, but on the whole we are not too convinced by Kalnins’ somewhat hasty treatment of ‘thousand’ (p. 35 f.).

Kalnins 2015: 56; it has been rejected by many though without a satisfying alternative.)

5.4.1. Hamp

Hamp 1973: 174 starts from **tuH-sk'-ont-* > **tuskont-* (for all three branches Baltic, Slavic and Germanic, which is impossible, 5.1.1), a form he rightly insists was not a compound of ‘hundred’ (but underwent its influence in Old Prussian, p. 172, and perhaps Germanic, p. 174).

It is not fully clear if Hamp sees **-sk'-* as inchoative or as an unexplained suffix. He writes: “We appear superficially on morphological grounds to have a present participle **tuskont-*, or **tuH-sk'-ont-*, of a base **teuH-* forming a **-sk'-* present in the expected zero-grade” (p. 174), which leaves it open whether what is mere appearance is the participial status of **-ont-*, the inchoative value of **-sk'-*, or both.

Hamp emphasizes that the word is an old ablauting formation, **-ont-/*-nt- > Balt *-ant-/*-int-* (p. 173 f.), which by the way is not compatible with an inchoative present participle.

Hamp was among the first to assume a double reflex of PIE **sk'* depending on whether it was followed by a back vowel (including *a*) or a front vowel. The Baltic reflex of **sk'* has variously been thought to be *š* (Brugmann 1886: 1897: 781, Gauthiot 1906: 4, and others), *šk* (Stang 1966: 92), *st* (Leumann 1942), *sk* and *st* (Pedersen 1943: 188-192, for Proto-Balto-Slavic), *st* and *š* (Hamp 1973: 173-175), then again *sk* and *st* (Gorbachov 2006: 11, 2014, for Proto-Baltic, followed by Kalnins 2015: 56 and ourself, below 5.5).

5.4.2. Comrie

For Comrie 1992: 792 f., an earlier **tūšant-* (< **tūsont-*) was reinterpreted as a present participle (perhaps ‘that which is fat (big)'), then a *t* was inserted by confusion of this present stem **tūša-* and the newer inchoative present **tū-sta-*.

5.4.3. Smoczyński

Smoczyński 2018: 1533 thinks of the verbal suffix *-st-a-*, mais celui-ci est propre au baltique, où il représente i.-e. **-sk/-e/o-*. Il part d'un participe présent **tuñk-st-ant-* ‘an increasing (hundred)’, mais **-nk-* aussi serait propre au baltique. On pourrait tout au plus comprendre ce **tunk-st-ant-* comme une réfection (étymologie populaire), c'est-à-dire supposer qu'un **tūšant-* hérité ait été réinterprété en baltique (oriental) comme un participe de présent inchoatif ou de changement d'état **tū-st-ant-*. Cependant, la motivation sémantique manque pour un verbe de changement d'état (pourquoi ‘a growing hundred’ plutôt que ‘a bid

hundred' ?), alors que $*s \rightarrow st$ représenterait une modification phonétique notable. De plus, cette explication ne rendrait pas compte du s slave, pour lequel une autre hypothèse resterait nécessaire (telle que l'influence du mot 'cent', comme nous l'avons vu).

The objection is of a semantic nature (5.4.3): 'thousand' will not be described as a change of state, 'a growing hundred', 'a growing amount', but in stative terms, 'a big hundred', 'a big amount'.

5.4.4. Our second attempt

On the other hand, the introduction of the *-sta-* suffix could be understood if it results from an accidental *reinterpretation* of a non-inchoative pre-form.

We suggest a hypercorrection. It can be speculated that the inchoative present participle suffix *-st-ant-* was frequent enough to undergo an occasional reduction to **-sant-* by dissimilation in fast speech. This may have led speakers to interpreting **tūsant-* 'thousand' as a careless pronunciation of **tūstant-* and to "restoring" that **tūstant-*. This process would have taken place before the assumed enlargement of the consonant stem **-ant-* to a vowel stem **-ant-i-*.

This is the solution I advocated in the Abstract sent to the organizers, before I knew Gorbachov's proposal..

5.4.5. Gorbachov

Yaroslav Gorbachov proposed an attractive solution in a 2006 talk, which he, however, has not published (except for his sound law, see c below), although there is a pdf of the handout. His solution for East Baltic can be summarized as follows (2006: 13)³⁶:

0) PIE $*túh_{xS}\text{-}nt\text{-}ih_2$ 'biggy' > $*tūs_{nt}ī$ as in Germanic and Slavic.

1) Folk-etymological alteration $*tūs_{int}ī \rightarrow *tūs\text{-}ćimtī$ after $*ćimta\text{-}$ 'hundred'³⁷. That would explain the appearance of the cluster $*sć$ (> st) even before any attraction of the (semantically unrelated) inchoative suffix. On the other hand, that would mean quite a big jump, unlike Slavic $*tūš_{int}ī \rightarrow *tūč_{int}ī$ (3.3.2).

2) $*sć > st$, as in the inchoative suffix PIE $*\text{-}s\acute{k}^e/o\text{-}$ but not only (PIE $*s\acute{k} > \text{Baltic } *sć > st$ before a *front* vowel: new sound law by Gorbachov, whose proof is sketched in 2006: 11 f. and detailed in 2014;

³⁶ In fact, Gorbachov assumes two varieties of East Baltic, the other being the extinct Baltic language that's reflected in the Finnic loanwords, but see 6 below.

³⁷ Gorbachov writes $*ć$ for Proto-Balto-Slavic and Proto-Baltic (from PIE $*\acute{k}$), while we posit PBS $*č$, PB $*š$ (from PIE $*\acute{k}$ and $*k_s$), but the difference is not relevant here.

3) Change of inflection: $*\bar{t} \rightarrow *-is$. Gorbachov sees two possible reasons (2006: 12):

a) Influence of *dešimtīs* (ibid., but the form need not be starred; already Lühr 1993: 124), even though the suffix was not as close ($*tūstantī$) as Gorbachov thought ($*tūstimtī$). Although ‘ten’ was originally a consonant stem, whose original inflection is still partially preserved (Endzelīns 1971: 182), it is likely that it shifted early to the *i*-inflection in the nominative singular (instead of expected $*dešins$) and in the plural cases with consonant-initial endings. (Even the dialectal gen. pl. $tūkstantū$ may be analogical of the archaism *dešimtī*.)

b) Pivotal role of accusative $*tūstimtin$. However, the accusative must have been in $*-jq$: Lith *sáldžiq*, Slav *bogynjq*, Goth *mauja*, 2.1.2.1 c, whereas Skt. *devīm* is analogical for $*devíyam < *daiwíHm$, no doubt after nom. *devī* and the type in nom. $-\bar{a}$, acc. $-\bar{a}m$.

Earlier (2006: 6), Gorbachov makes another remark:

c) “Historical *devī*-class is moribund in East Baltic: outside the fem. participles there are two *devī* nouns left in Stand. Lithuanian + two more in the dialects—all (...) designating persons (...).”

We retain *a* and *c* as plausible factors, not *b*.

4) $*-imtīs \rightarrow -antis$, perhaps by analogy of the participle of *-sta*-presents (2006: 12). This frees us of the necessity of positing a (late) PIE masculine (?) numeral $*tūsont-$ in addition to the feminine $*tūsantī$ of Germanic and Slavic. On the other hand, this involves two changes at once.

It would be more obvious to swap stages 3 and 4, so that $*-imtī$ still had $*\bar{t}$ when it underwent the analogy of $*-antī$. See also 5.5.2.1 below.

5.5. Our third attempt

There is a difficulty with Gorbachov’s solution³⁸. That $*-intī \rightarrow *-antī$ by attraction of the “normal” participle suffix (cf. 3.2) is possible, but, given the clear difference in Baltic between $*nt$ and $*mt$ (6.2), how can the influence of $*čimta-$ ‘hundred’ (with $*m$) on $*tūsintī$ (presumably with $*n$) winds up producing $*tūsčantī$ (with $*n$)? And after all, while in Slavic an influence of $*činta-$ on $*tūšintī$ to produce $*tūčintī$ (3.3.2) is but a small flick, in Baltic an influence of $*čimta-$ on $*tūsintī$ is less obvious.

5.5.1. Main changes

That’s why we finally opt for still a different solution: $*tūsintī \rightarrow *tūsantī \rightarrow *tūstantī$:

³⁸ Perhaps this is a reason why our colleague left his 19 year old talk unpublished.

5.5.1.1. *-int- → *ant-

Stage 1: Balt. **tūsintī* → **tūsantī* by attraction of the unmarked participial suffix.

That is, on the one hand, we gladly give up the earlier view that derives the East Baltic word from a masculine **tūsont-*, which is less economical, and prefer Gorbachov's suggestion of a reshaping on the model of participial **-antī*.

On the other hand, we do not follow our colleague as regards the timing and motives of this reshaping. Gorbachov places it at a late stage, as we have seen (5.4.5 e), and attributes it to a specific verb **tūk-sta-* allegedly reflected in Latv *tūkstu*, infinitive *tūkt* 'swell', which we don't deem possible. First, this suffixed present may well be a Latvian innovation only (cf. our fn. 27, which concurs with the "usual" view quoted in Gorbachov 2006: 12⁵). Second, the insertion of *k* in Lith *tūkstantis*, Latv *tūkstuotis* was only completed after the earliest literary documents (4.1), whereas the reshaping **i* → *a* took place much earlier (as it predates the pre-literary Latvian change **ant* > *ōt* > *uot*).

La préexistence d'un verbe n'est pas nécessaire pour cette réinterprétation (il suffit que l'attraction produise un verbe "virtuel"; cf. p. ex. *recrudescence*)

There's no need to worry about the lack of a specific trigger for this change **-intī* > **-antī*, because exactly the same happened in Slavic when **tysętji* → **tysq̃tji* (3.2). The change does not even need to have taken place earlier in Baltic than in Slavic, because **tysętji* → **tysq̃tji* must have taken place during or after the migration, thus roughly in the 7th or 8th century, while the earliest East Baltic literary text goes back to around 1500. The Slavic and Baltic changes are independent, however, because **tysętji* → **tysq̃tji* affected an area that may be described as South-West Slavic (3.2).

5.5.1.2. *-s- → *-st-

Stage 2: **tūsantī* → **tūstantī* by the (unusual) process that we have proposed above (5.4.4: formal attraction from inchoative **-stantī* made possible by a phonetic hypercorrection against the latter's fast-speech dissimilated variant **-santī*).

Here again, we do not attribute the change to a specific verb, but only to the high number of *-sta*-presents.

In our solution, there has been no attraction of 'thousand' by 'hundred' in Proto-Baltic and East Baltic.

5.5.2. Late changes

We happen to depart from Gorbachov's solution in our timing of the later stages as well: **tūstantī* → *tūstantis* → **tūkstantis*:

5.5.2.1. **-ī* → *-is*

Stage 3: change of inflection, **tūstantī* → **tūstantis*.

The factors are probably the following two (4.6 *a*, *c*):

1) The attraction of *dešimtīs* – even though the suffix was, by far, not as close (**tūstantī*) as Gorbachov thought (**tūstimtī*).

2) The **-ī*-inflection was almost only found in feminine participles.

Even so, the contradiction is surprising between our stage 2 (5.5.1.2, Gorbachov's stage 4), in which the word for 'thousand' is reinterpreted as a participle, and our stage 3 (Gorbachov's stage 3). A possible reason is that our word is likely to have increased its frequency, at least in its proper numeral function, between our stages 2 and 3: cf. Cleasby-Vigfusson's remarks on ON *þúsund* (above, 2.4). In early times, 'thousand' is likely to have been relatively unfrequent, and mostly used in its figurative meaning 'innumerable'; accordingly it fell prey more easily to the merely formal attraction of participles in **-antī*. Later, it presumably became more frequent, thus more independent (from the participles – hence the possibility of acc. **-in* entailing nom. *-is*), and more definitely felt as a numeral – which opened the way to an influence of *dešimtīs*³⁹. (The difference of accentual paradigm seems not to have been an obstacle.) Note that Gorbachov's different relative chronology does not allow this explanation of the contradiction.

5.5.2.2. *k*-insertion

Stage 4: As we have seen, an influence of the root(s) **tuk-*, **tūk-* is not likely (4.1.2), and, as we have seen as well (5.5.1.1, 4.1.2), the existence of a verb **tūksta-* is unlikely and unnecessary.

k-less forms are still attested in the oldest Lith and Latv documents (one each, 4.1.1).

³⁹ To tell the truth, we have not checked the frequency and semantic use of 'thousand' in Old Lithuanian texts. But note that the only example of 'thousand' in Old Prussian shows the "vague" meaning ('for thousand generations', 6.1.1).6.1

6. Old Prussian

6.1. Attestation

6.1.1. Form

The only attestation is the accusative plural *tūsimtons* (*Enchiridion* 37₁₈, Trautmann 1910: 29₇; 16th century): *en tūsimtons streipstoos* ‘in tausent Gelied, in thousand generations’. These are the only instances of spellings *-ons* or *-oos* for Baltic **-ans* (Trautmann 220).

6.1.2. Gender

The word is probably masculine, although a feminine is technically possible (because the accusative plural of *ā*-stems is generally spelt *-ans* like that of *a*-stems, see Trautmann 228, where all examples of § 134 a-b are from the *Enchiridion*).

The *Enchiridion* still has the neuter gender, although many neuters have already shifted to the masculine (Trautmann 1910: 219⁴⁰, Petit 2000: 32, 33 bottom, 2010: 153). We thus don’t know whether the masculine *tūsimtons* reflects an earlier neuter (like ‘hundred’) or not.

6.1.3. Number

The Prussian word is used in the plural (like Greek χῆλαιοι), unlike its East Baltic or Slavic counterparts (*tūkstantis*, *tysęšti*)⁴¹.

Although the *Enchiridion* is full of translation errors, there is no ground for amending this word. It’s unlikely that the final *s* of *tūsimtons* is due to anticipation of the initial or final *s* of **streipstons*, because Will generally had his informant translate only one word at a time (Petit 2022). Unlikely, too, that the informant had mixed up *tausend* ‘thousand’ (pronounced *tausent* which is Will’s spelling) with *tausende* ‘thousands’.

⁴⁰ The three Catechisms, of whom the *Enchiridion* is the third, are written in the dialect of Sambia (*Samland*), Trautmann 1910: XXI, 131.

⁴¹ Compare the use of ‘ten’ (Trautmann 1910: 320, 251 f.): *dessempsts*, *dessimpts* nom. sg. (other cases unknown, but no doubt a feminine *i*-stem) in the first two Catechisms, once *dessimton* (nom. sg. n.) in the *Enchiridion* (*Stai Dessimton Pallaipsai*), twice *dessimtons* (acc. pl.) in the *Enchiridion* as well (*Dessimtons Pallaipsans*, *stans Dessimtons pallaipsans*). – The neuter gender of *dessimton*, contrasting with the animate gender of *dessimpts*, must be analogical to **simton* which is the expected form of the word for ‘hundred’. The same holds for its shift to the thematic inflection.

6.2. The final

For Trautmann 1910: 252, an earlier **tūsintis*, with the same ending as Lith. *tūkstantis*, underwent the same fate as *dessimpts* ‘ten’; that is, the shortening of the *-is* nominative ending to *-s* induced a late shift from the *i-* to the *a-*inflection⁴².

Though late (postdating the syncope is **-as*, **-is*), this process must have taken place at a time when the word for ‘thousand’ was still a substantive (as in East Baltic or Proto-Slavic) and not yet an adjective (as in the Enchiridion, 6.1.3).

The change of inflection entailed a change in gender, from feminine to masculine, not easily attributable to ‘hundred’ which may still have been neuter.

6.3. The nasal

The *m* in OPr *tūsintons*, as opposed to *n* in Lith *tūkstantis*, is analogical to the word for ‘hundred’ (Hirt 1896: 347, Vaillant 1958: 647, Hamp 1973: 172, Lühr 1993: 129, Gorbachov 2006: 5, 13, Jakob 2024: 140³¹). This is the obvious explanation, even though ‘hundred’ is unattested and the two words may have been of different genders (6.1.2 and fn. 41).

The assumption (*etymologiae causa*) of a Proto-Baltic and Proto-Indo-European **mt* in ‘thousand’ (Brugmann 1911: 48 f., Endzelīns 183, Pokorny 1083, Lühr 1993: 124, 1998, 897, Jakob 2024: 140³¹) cannot be held. The clusters **-mt-* and **-nt-* are clearly distinguished in the Baltic languages (cf. Jakob l. c.), at least in the basic vocabulary, and ‘hundred’ had **-mt-*, ‘thousand’ had **nt-*. Compare on the one hand Lith *šimtas* ‘hundred’, lett. *simts*; lit. *dešimtis* ‘dix’ (et *dėšimt* indéclinable, forme réduite de l’accusatif), Latv *desmit*, OPr *dessimpts*, *dessempts* (f.) and *dessimton* (fn. 41), ordinal respectively *dešimtas*, *desmitais*, *dessimpts*; on the other hand e. g. *añtras*, *uōtrs*, *antars* ‘second’, infixed presents like Lith *kriñta*, Latv *krīt* ‘he falls’ (Endzelīns 1971: 216), numerals *deviñtas*, *devītais*, *newints* ‘ninth’⁴³ and, precisely, Lith *tūkstantis*, Latv *tūkstuots*.

Neither is there a reason to favour a spontaneous phonetic change. Since there exists a sporadic dialectal dissimilation of *nt*, *nd*, *nk* to *mt*, *md*, *mk* in Lithuanian and Latvian (Smoczyński 2018: passim⁴⁴) – often after *u* or at a distance

⁴² No influence of **simta-* ‘hundred’ is expected here, as the latter was presumably still neuter.

⁴³ The apparent exception *septiñtas*, *septūt(ai)s* does not reflect a PIE **septmtos*, which did not exist, but it replaces Lith *sėkmas*, OPr *sep(t)mas* < i.-e. **septmos* (cf. Endzelīns 184 and OCS *sedmъ*). Like the cardinal Lith *septyni*, Latv *septiņi*, the new ordinal is derived from **septin* < **septm* (Comrie 1992: 756). The treatment of final **m* is independent from that of *m* before *t*.

⁴⁴ Pp. 230 *dimsta*, 245 *drēmžti*, 297 *firānka*, 381 f. *grēmžti*, *grēsti*, 405 *gūndyti*, 546 *kinkýti*, 609²⁶⁷ (*krīsti*: various exx.), 688 *lentà*, 1011 *praklėntas*, 1063 *rantýti*, 1082⁽⁴³⁶⁾ *rémžti* and others, 1174 *siūndyti*, 1316⁵⁰⁸ *wissambs*, 1317 *stūnda*, 1381 f.⁽⁵²⁶⁾ (various), 1383 *šiñkorius*, 1537 *tūntas*. – See also Fraenkel 1953: 107 f.

after *p, f* – Smoczyński attempts to explain in that way the *m* in Prussian *tūsimtons* (l. c. 609²⁶⁷, 1316⁵⁰⁸, 1533) and even in the Baltic words for ‘ten’ (tacitly p. 216) and ‘hundred’ (XLI, 1011, 1381 f.). That’s however impossible for the latter two, in which *mt* is not sporadic but pan-Baltic and constant (and is even generally seen as going back to Proto-Indo-European). And that’s unlikely for *tūsimtons*, because *nt* > *mt* is not attested in Old Prussian and the analogy of ‘hundred’ yields a sufficient explanation.

6.4. The stem

Even not counting the final (*a*-stem, 6.2) and the nasal (*m* for *n*, 6.3), the Prussian form *tūsimta-* is markedly different from East Baltic, Lith *tūkstantis*.

(a) One way of accounting for all the remaining differences at once is put forward by Smoczyński 2018: 1532 f. s. v. *tūkstantis*, who sees a loan from MHG *tūsent* as “most likely”. As is known, both Low German and High German were present in Prussia (see e. g. König–Paul 1994: 64, 74 f., 184, 188).

The borrowing may even have been from New High German, because the German of the Elbing Vocabulary (14th or 13th century, Petit 2010: 22) is precisely a dialect of High German⁴⁵ that had not diphthongized MHG *ī, ū, ū̄*⁴⁶ even though it shares in the monophthongization of *ie, üe, uo*⁴⁷. The word for ‘thousand’ itself, as we have seen, is not attested before the 16th century (A question could be why Old Prussian would have rendered the German *ə* of the second syllable as *i*, but it’s hard to tell what the “expected” vowel would have been anyway).

(b) On the other hand, “deep” differences between East and West Baltic in some features are not unknown, especially in the field of numerals: compare the words for ‘third’, OPr *tīrts* < **tritas* (metathesis, Smoczyński 2018: 1507)⁴⁸ vs.

⁴⁵ E. g. 16 Woche, 17 Sontag, 30 Tal, 41 Hiczcz, 76 Scheitel, 86 Naseloch, 92 Czan, 94 Czung, 119 Citzcz, 120 Hercz, 210 Tho^ere, 211 Grosthor, 212 Phorte, 214 Rochloch. Exception: 100 Schulder.

⁴⁶ E. g. 34 Vu^eer, 56 Is, 95 Gume, 111 Mus, 114 Dume, 188 Wip and the unambiguously High German 11 Tufel, 57 Ryf, 61 Tych, 122 Buch (‘belly’). In e. g. 113 Vu^est, 193 Hu^es, 802 Czuⁿ, superscript *e* presumably spells length rather than umlaut, cf. fn. 47 and 800 Styeg. Exception: 120 Seyte ‘side’ with diphthong.

⁴⁷ E. g. 43 Glut, 62 Vlys (‘stream’), 63 Reynvlys (‘rain fall’), 128 Nyre, 139 Dy, 140 Kny, 160 Blut, 170 Muter, 173 Brud(er), 180 Stifmut(er), 217 Barkenstul. In e. g. 145 Vues, 146 Vu^essale, 148 Vu^esbret, 216 Stu^el, *ue* or *u^e* is thus presumably a historical spelling for long *u*, cf. fn. 46.

⁴⁸ There is no inherited PIE **tīt-* either in OPr *tīrts* or in Old (non-Dardic?) Indo-Aryan *tītīya-* < **trītīya-* (by dissimilation: Osthoff 1881....., Milizia, diss. 2003 + JIES 2004....., cf. *śṛṇu-* pres. ‘hear’ < *śṛṇu-*, ll. cc. ... and Scheftelowitz, KZ 1925, « Ein urindisches Liquidengesetz », spec. p. 262-265). Far from the combinations by G. Schmidt, « Idg. Ordinalzahlen » (IF 97, 1992), there are thus no other PIE forms for ‘third’ to be reconstructed than **tr(i)yo-* (not directly

East Baltic *trēčias*, *trešs* < **tretjas*, Slavic **tretiъ* < **tretijas* (< **tritiyos*); ‘six, sixth’, OPr **uscha-*, *usts* < **(š)uš-ja-*, **(š)uš-ta-* vs. Lith *šeši*, *šėštas*, Sl *šestъ*, *šestъ* (Viredaz 1997: 116, with reff. fn. 23 to Bezzenberger for the OPr cardinal); or more modestly ‘nine’, with OPr *n-* vs. East Baltic and Slavic *d-*.

Since we have been led to posit **tūsintī* for an early stage of East Baltic (5.5.1.1), another solution is thus to derive the Old Prussian form directly from it⁴⁹ (plus the late changes discussed in 6.3, 6.2).

(c) Variants of scenario (b) cannot be excluded because OPr /s/ can in principle reflect any of the three BSl phonemes **s*, **š* and **č*, a pre-merger **s* without RUKI is slightly more likely than **š* with it, because of the generally closer relationship with East Baltic than with Slavic; and a alteration of the sibilant under the influence of ‘hundred’ is less likely than in Slavic because of the greater difference between ‘hundred’ and ‘thousand’⁵⁰.

In conclusion, solution (b) seems to be the most likely although (a) or the alternatives mentioned in (c) are perhaps not excluded.

6.5. Conclusion (West Baltic)

Perhaps the OPr word was borrowed from a regional High German **tūsant* (6.4 a), as **tūsintas* or *-an*, with *i* for German *a*, then altered to **tūsintan* after **simtan* ‘hundred’ (6.3) and finally to *tūsims** in later Old Prussian (Enchiridion) in the process of loss of the neuter gender. But maybe it could be why *a* was rendered by *i* rather than another vowel, or whether there had been enough time for *nt* to be altered to *mt*.

Or perhaps West Baltic directly reflects Proto-Baltic **tūsintī*, unlike East Baltic that reshaped it to **tūsantī* (6.4 b, 5.5.1.1). Later, on the other hand, **-nt-* was changed to **-mt-* after (**čimtan* >) **simtan* ‘hundred’ (6.3), the word shifted to

attested, but cf. I-Ir **ktur(i)ya-* ‘fourth’ and Szemerényi, *Numerals ...*, **tri-to-* (regular) and **tri-t-(i)yo-* (a blend of the two others).

⁴⁹ This is in fact, mutatis mutandis, Trautmann’s explanation (6.2).

⁵⁰ In this paper and especially in this paragraph 6.4, it is assumed that PIE **k̑* and **s* were treated in Old Prussian in the same way as in Lithuanian before **š* merged with *s* in the former. That is: (1) RUKI only took place sporadically after **i*, **u* (4.2); (2) PIE **k̑* > Balto-Slavic **č* > Baltic **š* > Old Prussian *s* though a late merger (as in Latvian). This is the most economical solution. Alternatives are thinkable:

(1) RUKI did not take place at all after **i*, **u* in West Baltic.

(2) RUKI took place throughout after **i*, **u* in West Baltic (as in Slavic).

(3) PIE **k̑* > BS **č* > OPr *s* due to an early change **č* > **c* as in Slavic.

In our opinion, (1) and (2) are unlikely and (3) is not worth considering. If, nonetheless, one or two of them are retained, it’s easy to rewrite our 6.4 accordingly.

the (feminine) *ī*-declension (as in East Baltic, 5.4.5:3, 5.5.2.1) and finally to the (masculine) *a*-declension (6.2).

7. Fenno-Volgaic languages

7.1. Forms

Our word is also found in Fenno-Volgaic or North-Western Uralic languages, namely Fennic⁵¹, Sámi, Mordvin and Mari⁵², presumably as direct and indirect loan from East Baltic given the *a* of the second syllable: Finnish *tuhat*, gen. *tuhannen*; Estonian *tuhat*, gen. *tuhanden*; Sámi *tuhāt*⁵³; Mordvin *t'ožań*; Mari *tüžem*⁵⁴.

7.2. *u* vs. **ū*

7.2.1. Problem statement

The shortening of old **ī*, **ū* is regular in Fennic (cf. most recently Pystynen 2022: 12-14), so that it could account for the short *u* in *tuhat* if the borrowing took place early enough.

Jakob 2024: 108 f. doubts that this is the case. Indeed, unlike those from Germanic, most borrowings from Baltic are later than the shortening in question and reflect Baltic *ū* as *uu* (l. c. 108, citing Plöger 1982: 93)⁵⁵. According to Jakob, there is not even a single convincing example of **ū* > *u* in Baltic loans.

⁵¹ Or Finnic, Baltic Finnic, French fennique, German Ostseefinnisch. – We prefer Fennic over Finnic because it allows us to use Finn as unambiguous abbreviation for ‘Finnish’.

⁵² We use “Fenno-Volgaic” as a cover term for these four branches, without any phylogenetic implications. The same goes for “Volgaic”, even though Mordvin and Mari do not form a phylogenetic “clade” or “node”. – At first we wanted to keep the Fenno-Volgaic section very short, but the differences between early East Baltic **tūsantī* and early Fennic **tušante* decided otherwise.

⁵³ *tuhāt*, *duhat*, *dufat*, *doḡāt*, *tōvańt* etc., see *SKES* s. v. *tuhat*. These are recent loans from Finnish, later than the Finnish changes **š* > **x* (> *h*) and, for most, **-nt* > *t*. – Add Swedish Sámi *tusan* from Swedish (Thomsen 1890-1893: 233).

⁵⁴ Thanks to Anthony Jakob and Juho Pystynen for their bibliographical and other help in relation with the Finno-Ugric part.

⁵⁵ Pystynen, l. c. 12, has a list of 11 loans from Germanic (and a few others) that shortened their **ī*, **ū* and Plöger, l. c. 94. a list of 5 others that did not. It may be presumed that the former are earlier than the latter, but I cannot confirm this by semantic or other criteria. – In loans from Baltic, Plöger l. c. 93 f. cites three examples of **ī* → *ii*, two of **ū* > *uu*, one of closed **ē* (from **ai*) > *ii*, two of **ō* > *uu*.

7.2.2. *kulo*

7.2.2.1. The question

Thus, writes Jakob, although Finn *kulo* ‘wildfire; last year’s grass’ certainly corresponds to Latv. *kūla* ‘last year’s grass’, the etymology of the Baltic word is uncertain and so is the direction of the borrowing (Jakob 2024: 143, w. lit.). Similarly, *kulo* is not listed in Suhonen 1988, although it is in Pystynen 2022: 12.

Yet the Indo-European etymology may be better than is deemed by *ALEW* (s. v. *kūlē*) and Jakob (l. c. with fn. 37)⁵⁶.

The relation between Greek καίω ‘kindle’, med. ‘burn’, aor. ἔκηα and ἔκαυσα < *kh₂w-ye/o-, *keh₂u-s- and the anteconsonantal zero grade *kh₂u- > *kuh₂- > *kū- in Lith *kūlės* ‘Brandpilze’ (*LIV*² 345) and Latv *kūla*, Lith *kūlymas* ‘last year’s grass’ (Smoczyński 2018: 625) is not only “formally possible” (Jakob l. c.), but it seems semantically sound (see below⁵⁷). It would have been a weakness if the root were only found in two branches, Greek and Baltic, but a likely third witness is Toch A *koṃ*, B *kaum* ‘sun, day’ < *kāu-ni- < *keh₂u-ni- (cf. *ALEW* online s. v. *kūlē*, Adams 2013: I 225).

7.2.2.2. The Finnish material

Finnish *kulo* has a double meaning. On the one hand, ‘withered grass, last year’s grass’ (Thomsen 1890-1893: 11¹, 99, 122, 144, 190 f.) , ‘longue herbe desséchée et jaunie par l’hiver’ (Yrjö-Koskinen 1900), ‘ungemäht gebliebenes vertrocknetes Gras vom Vorjahr’ (*SES*); cf. Karelian *kulo* ‘sun-scorched hay (usually from the previous year) left uncut or mowed too late’ (*SES*).

On the other hand it means the same as *kulovalkea* (Y.-K., *SES*, perh. Thomsen 190) (with *valkea* ‘white, bright’ and ‘fire’, though ‘fire’ is usually *tuli*), namely: ‘Lauffeuer, Waldbrand’ (Thomsen 190), ‘incendie d’une forêt, d’une prairie, traînée de feu’ (Y.-K.), ‘Lauffeuer’ (*SES*), that is, ‘grass fire, uncontrolled fire in a dry grass area or, by extension, a forest’. Hence also Finn *kulouta*, *kuloutua* ‘mit vorjährigem Grase bedeckt werden, verbrannt werden’, *kulottaa* ‘vorjähriges Gras anzünden, Waldbrand machen’ (Thomsen l. c.).

Other compounds include: *kuloniitty* (*niitty* ‘meadow’), ‘pré dont on ne fauche l’herbe que tous les deux ans, a meadow that’s only mowed every two years’ (Y.-K.); *kulonalainen* (*ala* ‘under’) ‘qui vit dans l’herbe desséchée, qu’on n’a pas fauchée’.

⁵⁶ The *SES* entry *kulo* even has an addition (KR 2021): “Nowadays, its Baltic origin is considered certain.” However, as we have seen, this has now been challenged again by Jakob l. c.

⁵⁷ Our English translations from Finnish (*SES*), Danish (Thomsen) or even German are tentative.

Derivatives include: Finn *kulouta*, *kuloutua* ‘mit vorjährigem Grase bedeckt werden, verbrannt werden’, *kulottaa* ‘vorjähriges Gras anzünden, Waldbrand machen’ (Thomsen l. c.);

7.2.2.3. The Baltic material

– **kūlā* in Latv *kūla* ‘last year’s grass, dry grass; moulted fur, snake’s skin’ (Smoczyński 2018 s. v. *kūlēs*; ‘altes dürres Gras, namentlich vom vorigen Jahre übrig gebliebenes; das alte Haar des Viehs zur Zeit des Abwerfens’, Thomsen 1890–1893: 190);

– Latv *wilka-kūla* ‘ganz schlechtes Morastgras. welches das Vieh nicht frisst’ (Thomsen); Lith (Memel) *kules* ‘eine Grasart auf Torfwiesen’ (ibid.);

– Lith *kūlymas*, *kūlimas* ‘last year’s grass that has not been cut’ (Smoczyński l. c.);

– Lith *iš-kūlyti* ‘to dry, wither’, which, as Jakob notes l. c. fn. 13, cannot be related to its Latvian synonym *kālst*, root *kālt-*;

– Latv “*kūlāt* (*kūlēt* with multiple forms) ‘dürre werden’” (Thomsen l. c.);

– Latv *kūlas uguns*, *kūlas ugunsgrēks* ‘dry grass fire’ (*uguns* ‘fire’, *grēks* ‘sin, transgression’, *ugunsgrēks* ‘(uncontrolled) fire’. Judging by the Google results (Sept 2025), the phenomenon seems to be very common and accidental.

– Lith *kūlēs* f. pl. ‘Brand im Getreide’ (Thomsen), ‘stem rust, parasitic fungi causing dark spots on stems, ears, grain, *Ustilago*; Brandpilze, Staubbbrand, zgorzel’ (Smoczyński 2018: 625, w. lit.), ‘charbon des céréales’; with metatony before -ė, cf. Derksen 1996: 54-58, 62 f.; from **kūl-ijā-*; also *kūlēti* ‘brandig werden’ (Thomsen), *kūletas* ‘infected with stem rust’. Other Lithuanian names for *Ustilago* are *degulys* (from *dėgti* ‘burn’) and *súodžiai* (‘soot’) (Smoczyński 2018: 877, 1328).

7.2.2.4. Discussion

The majority of items in both the Fennic list and the Baltic list above relate directly to ‘dry grass’, not to ‘grass fire’ or ‘burnt grass’. This suggests that the adjective **kū-ló-* underlying Balt **kūlā* did not mean ‘burnt’ (thus Walde-Pokorny 377) but ‘inflammable’ Withered or dry grass is highly inflammable and grass fire spreads rapidly. It thus made sense to use a word meaning ‘inflammable’ to describe the ‘dry’ stage reached by grass or grassland areas. Later, the meaning could be extended to the withering state of other plants as well.

Is an adjective **kū-ló-* ‘inflammable’ plausible from the IE point of view? See Brugmann 1906: 373-377, Endzelīns 1971: 109 f., Chantraine 1933: 237-251 for the (multiple) semantic values of PIE, Baltic and Greek derivatives in *-*lo-*, -*la-*, -*λο-*. “Inflammable” could be classified among the adjectives or nouns

expressing the “Begriff einer tadelhaften Gewohnheit” (Brugmann: 374), “some kind of negative characteristic” (Endzelīns: 109 f.) or having a “valeur ‘péjorative’” (Chantraine: (238-)239), like Latin *bibulus*, *crēdulus*, *gemulus*, *querulus*, OHG *ëzzal* ‘voracious’, Lith *kūšlas* ‘small, puny (etc.)’ (< *‘tired’, Smoczyński 2018 s. v., or perhaps rather *‘easily tired’), Latv *mirla* f. ‘miserable, emaciated creature’, Greek δειλός ‘coward’ (Il. cc.). A PIE **kh₂u-ló-* (> **kū-ló-*) *‘easily burning, having an unpleasant tendency to take fire, inflammable’, would probably fit in the same frame – provided the basic meaning of the root was not ‘kindle’ (transitive, LIV² 345 ‘anzünden, verbrennen’) but ‘burn’ (intransitive). This is indeed in line with the Tocharian derivative *koṃ/kaum* ‘sun, day’ (7.2.2.1), but requires that the Greek transitive καίω be secondarily derived from the medio-passive καίομαι, a process that is well known in Greek (cf. φθείρω, ὀλλῶμι, σφάλλω), but generally not obvious from the Greek data alone.

I’m not sure if some Baltic or Fennic derivatives require that the adjective **kū-lo-* was polysemic and meant both ‘inflammable’ and ‘burning’ or ‘burnt’, or if the whole word family can derive from ‘inflammable’, ‘dry grass’. Thus, Finn *kulottaa* ‘put fire to dry grass, start a grass fire, a forest fire’ could have originated as ‘to do dry grass’, i. e. to do what is usually done to it, to take care of it; Finn *kulo* in the sense of ‘grass fire, wildfire in grassland or forest’ could have arisen either as shortening of compounds *kulovalkea*, *kulotuli* (the latter used in Estonian) or by metonymical extension of ‘dry grass’. The latter would entail that the same word could mean both ‘dry grass’ and ‘burnt grass’, which seems unlikely. Lith *kūlės* can hardly be the *testis unus* for an early sense ‘burnt’ of **kūla-*. Perhaps, unlike many of its synonyms in various languages, this name of the disease does not derive from ‘burnt’ but from ‘withered, dried’⁵⁸.

7.2.2.5. Conclusion (*kulo*)

Independently of the etymology, there may be a few indications for the direction of borrowing to be Baltic > Fennic: it seems easier to explain a shortening of **ū* in Fennic than a lengthening of **u* in Baltic (7.2.2.1); ‘dry grass’ is close to semantic fields such as agriculture (e. g. ‘hay’, Finn *heinä*) or soil conditions (e. g. frost) that have a good number of Baltic-to-Fennic loans but perhaps none in the other direction (I haven’t elaborated on this point above).

The etymology is a strong indication because no etymology of Finn *kulo* is known on the Uralic side while the Indo-European etymology of Balt **kūlā* ‘dry grass’ seems rather sound (7.2.2.1, 7.2.2.4).

⁵⁸ Or from *‘dried out and liable to get burnt (like dry grass)’, but that seems too subtle.

7.2.3. *lyhyt*

Another possible example of shortening is Finnish *lyhyt* ‘short’ (gen. *lyhyen*), if from the Baltic feminine active past participle **lūž-us-ī* (Lith *lūžusi*, of *lūžti*, intransitive) ‘broken’ (Liukkonen 1994, Pystynen 2022: 12)⁵⁹, or more probably from the masculine accusative **-usin* (Lith *-usi*)⁶⁰. We are not aware of another etymology for *lyhyt*⁶¹.

For the final *s*, Liukkonen p. 2 compares *airut* ‘herald, messenger’ < Germanic **airuz*; *olut* ‘beer’ ~ Lith *alus*; *ohut* ‘thin’ ~ Latv *ass* ‘pointed, sharp’ < **ašus*. This does not seem to be a regular change, however, since nominatives in *-us*, *-ys* do occur.

Liukkonen l. c. attributes the shortening of **ū* to the position before *h*, for which he adduces not only *tuhāt* but also *puhdas* (below).

7.2.4. *puhdas*

Finnish *puhdas* ‘clean, clear, pure, sheer (unadorned)’ (gen. *puhtaan*) is usually traced to Indo-European **puHtos* (Mayrhofer II 106, *SES* s. v.) or better Indo-Iranian **puHtas* (Wiktionary s. v., accessed 9.25). However, that makes for an uncannily early borrowing for a word that’s only attested in Baltic Fennic.

Liukkonen 1994: 91 has a more plausible proposal in this respect, the Baltic participle **pōštas*⁶². The Baltic verb is Lith *puošti*, dial. also *púošti*, ‘decorate, ornament, embellish, beautify’, Latv *puost* ‘clean, sweep, clear; decorate, beautify’ (Smoczyński 2018: 1038). The semantics fits rather well. According to Liukkonen, the word was borrowed in Fennic as **pūštas*, hence **pūhδas* > **puhδas* > *puhdas*⁶³.

7.2.5. Migratory argument

Let us add that, for geographical reasons, the first contacts between Fennic and Baltic must have been even older than those with Germanic, so that the

⁵⁹ Not retained in *SES* or Wiktionary (fn. 61).

⁶⁰ See Endzelīns 1971: 257 for the inflection of the active past participle. – As for the Lith root *lūž-/lauž-* ‘to break’, its own etymology is complicated by the fact that it implies an Indo-European palatal, whereas its Indo-Iranian cognate implies a velar (*LIV*² 415 f. and n. 1); nevertheless, it is not disputed; perhaps the languages of Europe have merged **leug-* ‘to break’ and **leug-* ‘to bend’.

⁶¹ Wiktionary in English speaks of an “unknown root” (<https://en.wiktionary.org/wiki/Reconstruction:Proto-Finnic/1%C3%BCh%C3%BCt> 19.06.2025). Versions in other languages do not give any etymology. *SES* (accessed 09.25) derives it from an ad hoc root.

⁶² *Puhdas* would thus be added to Plöger’s examples, cf. fn. 55.

⁶³ Liukkonen’s etymology is mentioned as a possibility in *SES* (in the bibliographical note, after Koivulehto).

hypothesis of a Fennic shortening of **ī*, **ū* in some of the Baltic loanwords is not as *ad hoc* as Jakob (2024: 108) fears.

7.2.6. Conclusion

In short, apart from *tuhāt*, the potential evidence for the shortening of Baltic **ū* in some Finnish loans is scarce, but not bad: we have only noticed two examples, namely *kulo* ‘dry grass, dry grass fire’ (7.2.2), whose Baltic source **kūlā* would have a good IE etymology, and *lyhyt* ‘short’ (7.2.2.5), from the Baltic participle **lūžusī* (nom. fem.) or **-in* (acc. masc.) ‘broken’.

For Liukkonen, the shortening of **ū* is simply due to the following *h*, for which he adds a further example, *puhdas* ‘pure’ < Balt **pōštas* ‘beautified’ (7.2.4). The rule would not cover *kulo*, so that we would still only have two parallels for the shortening in *tuhāt*.

It may be speculated that the examples of Balt. *ū*, **ō* > Finn *u* (the four listed here) are earlier borrowings than those of Balt *ū*, **ō*, *ī*, **ē* > Finn *uu*, *ii* (the four + four listed by Plöger 1982: 93). However, no such difference is obvious from the semantics of the two groups of loans.

On the other hand, alternatives to a Baltic origin of *tuhāt* are unpromising. A Germanic source would not explain the *a* in our opinion (2.1.2.3). A loan from Slavic (**týsešti*, i. e. **tūsEntjī*) would not account for the Fennic form at all. A loan from an unknown, non-Indo-European language is, in our opinion, too “big” a hypothesis for such a “small” difference as Fennic **uš* vs. Baltic **ūs*; it would not really explain the Fennic form since the sound laws of that language are not known; and it would not explain the Germanic-Baltic-Slavic word itself (see also 5.3, 8.1).

All in all, our position is that Finnish *tuhāt* is best explained as a loan from Baltic and that the short *u* is not an obstacle to this, whatever its cause.

7.3. **š* vs. **s*

7.3.1. Baltic **š*?

Finnish *h* in *tuhāt*, from a former **š*⁶⁴, is generally seen as evidence that the Baltic source language had **š* due to the RUKI law. This is undoubtedly the simpler hypothesis from the Uralic point of view. However, this is not what the Baltic and IE evidence points to:

– Lith *tūkstantis* and its variants are always attested with *-kst-* in Lithuanian; the only exception of Bretkūnas is not an archaism (4.2);

⁶⁴ Cf. **š* > *x* in e.g. Slavic, Spanish, perhaps for different reasons.

– Many Baltic words do not have **s* > **š* after *i*, *u*, and this *is* an archaism (4.2, after Stang).

Now we cannot exclude that the Fennic word was borrowed from an aberrant Baltic dialect that had **tūšantī* (or **tūštantī*) with **š*. Nevertheless, this is uneconomical (from the Baltic point of view). Although the Baltic loans in Fennic must indeed be from an extinct Baltic dialect (Nieminen 1957), that dialect is unlikely to have had **š* where Lithuanian does not⁶⁵. (If **s* > **š* is an innovation that failed to reach Baltic because of its marginal position, it is not expected to have reached an even more marginal area.) We will thus explore a different solution (7.3.2-7.3.4).

7.3.2. **š* from foreign **s*

It is not uncommon for a Finnish *h* (< **š*) to reflect an **s* of the source language, whether Germanic (Koivulehto 1984: 115-117, Kallio 2015: 6 f.), Slavic (e.g. Finn *Pihkova* ‘Pskov’ < *Pbskovъ*, Old Russian *Plěskovъ*), or Baltic (Jakob 1924 b: 142), even after vowels other than *i*, *u*. However, most of these cases are not very relevant to the one at hand:

– Germanic, having no opposition between *s* and *š*, must have had an apical pronunciation of *s*, which sounds midway between *s* and *š* to speakers who have both phonemes in their language and pronounce the former as “predorsal” or laminal (cf. Martinet 1955, 235 f., Koivulehto 1984: 115 f.);

– Loans from Slavic certainly belong to a later period than the Baltic borrowings, at least those Baltic borrowings that are old enough to have undergone the shortening of **ī*, **ū*.

Nonetheless, we have to provide examples of Baltic (Lithuanian) *s* vs. Finnish *h* if we are to make Balt **tūšantī* > *tuhat* plausible. This is what we’re attempting below (7.3.3).

7.3.3. **š* from Baltic **s*

A list of Baltic loans in Fennic (arranged by semantic groups) is given by Suhonen 1988⁶⁶. While loanwords with *h* vs. Lith *š* or *ž* are numerous (more than 30), those with *h* vs. Lith *s* are indeed very rare⁶⁷:

⁶⁵ It is safe to admit that Latvian formerly had **š* in the same conditions as Lithuanian does, either from PIE **k̑*, **k̑s* or from PIE **s*, before it merged it with *s* (like Mazurian Polish dialects) and acquired a new *š* from **č* < **tj*. Alternatives assuming that PIE **k̑*, **k̑s* reached Latv *s* through the same path as in Slavic, or that Latvian did not undergo RUKI at all, are more complicated and will not be considered.

⁶⁶ An earlier list, alphabetical, is found in the second part of Thomsen 1890-1893. We have not used it systematically.

7.3.3.1. Word-final examples

– Some words have *-e* < **-ex* < **-ix* < **-iš* vs. Baltic *-is*: at least *herne* ‘pea’, *helle*⁶⁸ ‘hot weather’, Lith *žirnis* ‘pea’, *šiltis* ‘Wärme, heat’.

These two are very important for our purpose, because these Baltic words are not *i*-stems, but *ia*-stems, with nom. sg. *-is* < **-īs* < **-ijas*, thus with no application of the RUKI law even if it had been fully active in the Baltic dialect concerned.

None of the Baltic loans ends in *-is*. Those in *-i* (like *hanhi* below) presumably reflect the Baltic accusative⁶⁹.

In addition, the contrast between Finnish **-eh* from Baltic **-īs* and Finnish *-as*, *-äs* from Baltic **-as* shows that word-final position is not the cause for the Finnish reflex **š*, or not its primary cause.

Unfortunately, these examples only concern Baltic **s* after **i*, not after **u*.

7.3.3.2. Word-internal examples

– *laiha* ‘thin, skinny, Lith *liesas* ‘id’. A Germanic **laisa-* (Jakob 2024: 142) does not exist, but only **laisizan-* ‘less’, which, though considered as related (Kroonen 2013 s.v.), is not as good a semantic match anyway.

– *vihvilä* ‘rush, *Juncus*’, Lith *vikšvā* ‘Riedgrad, Segge, sedge, *Carex*’, dial. *vikšvā* (probably from **wiswā*, cf. 4.1.1 above and Smoczyński 2018, s. v.). Possible cognates with different meanings in Slavic and Germanic (Fraenkel, Smoczyński).

– *hanhi* ‘goose’, Lith *žqsis* ‘id.’ (perhaps an assimilation **š-s* > **š-š*).

– *pahla* ‘fishing rod’, Lith *basl̥ys* ‘pole’. An apparently Lithuanian-only creation, indirectly on the root *bed-* ‘poke, pierce, stick’ (Smoczyński 2018, s. v.).

– *lahto* ‘bird snare in treetop’, Lith *slāstai*, *slāstos* etc. (many variants, though none with *š*) ‘(rat) trap’, Latv *slasts*, *sla(g)zds* ‘bird trap’⁷⁰. Etymology uncertain (Jakob 2024: 141, w. lit.).

– *mähkä* ‘Geschwür, Knollen (am Baum), Knoten’, Lith *mēzgas*, *māzgas* ‘Knoten, zugezogene Schlinge’ (Suhonen 609, with source; not found with this meaning in *SES*).

⁶⁷ Our sources for most of the etymologies were wiktionary.com and *SES*, where a few of those retained by Suhonen have been left out. – The Baltic source of the loanwords may have been closer to Latvian, but we cite Lithuanian for the preserved distinction *s* : *š*.

⁶⁸ The regular Finnish lenition of the initial stop of a closed syllable yields opposite results in paradigms in *-i*, gen. *-en* and in **-eh*, gen. **-ehen*, e. g. *joki*, *joen* ‘lake’ but *helle*, *helteen*.

⁶⁹ For the loss of Baltic final **n* in Fennic loans, compare *paimen* ‘shepherd’ corresponding to Lith acc. *piemenį* rather than nom. *piemuō*.

⁷⁰ Etymology and glosses after *SES*; details of Baltic forms after Smoczyński 2018.

– *ahingas* ‘Fischgabel, leister, pronged fishing spear’, ‘lysterjærn, iron spike for the leister’s prongs’, Lith *ākstinas* ‘thorn, prickle; sharp stick, goad’ (Suhonen 1988, 599, with reservation; Thomsen 1890-1893: 75, 82, 158; details of the Lith word after Smoczyński s. v.). Lith *ākstinas* and *akstīs* ‘small pole with a sharp end (etc.)’ are themselves borrowed from Slavic **ostъnъ* ‘spike, goad’ (Thomsen 158) and **ostъ* ‘spike, thorn (etc.)’. Given the IE source **ak-* or **aks-*, the Slavic origin of the Baltic words is proved by their having *s*, not **š*. (The variant *akštīs* might be inherited but it’s rather a secondary alteration of *akstīs*.) The Fennic word shows a change of suffix (Thomsen 82), but also an unexpected reflex of Balt *kst* or **st*. Thomsen 75 sees it as a case of *h* for expected *hd*, but the only other example cited is *tuhāt*, which cannot be from **hd* in view of the Mordvin and Mari *ž* (7.1, 7.7). Nevertheless, the change **hδ > h* is regular in other Fennic languages, cf. Finnish *puhdas*, Ingrian, Karelian, Estonian *puhas*, Votic *puhaz* (SES), so that its sporadic application in Finnish seems possible and would account for *ahingas*.

Though about half of these examples have a cluster **s* + consonant, this is hardly relevant because Finnish usually allows these clusters.

7.3.3.3. Counter-examples

Words in which Balt **s* (or **š*) corresponds to Finn *s* even after *i*, *u*, *r*:

– *karsas* ‘crooked (etc.)’, cf. Lith *(s)keĩsas* ‘transverse, sideways, crooked’. Slavic **čersъ* ‘through’ with no RUKI effect points to BS **(s)kertsā-*. At the time of the borrowing, **ts > s* had probably already taken place, like **č* (from PIE **k̑*) $>$ *š*.

– *laiska* ‘lazy’, cf. Latv *laĩšks* ‘id.’, but the etymology and direction of borrowing are uncertain.

– *kausta* ‘Seitenholz des Arbeitsschlittens’, cf. Latvian *skāusts* ‘wedge (?); neck, shoulder or loin (of an animal) (?)’⁷¹. A semantic link is not obvious. We don’t know the Baltic word’s etymology, nor whether it had **s* or **š*.

– *musta* ‘black’; Baltic etymology unlikely⁷².

⁷¹ Finnish word’s translation and Latvian word after SES; Latvian word’s translation after SES (‘niska, säkä’) and Ulmann 1872: 260 (‘der Keil, das Kreuz, der Nacken eines Thieres’. Not in Thomsen 1890-1893.

⁷² This last example, not listed by Suhonen (1988: 608, Farbenbezeichnungen), is adduced by Liukkonen 1994: 91, who derives Finn *musta* ‘black’ from a Baltic participle **musta-* ‘beaten (e. g. eye)’, cf. Lith *mūšti* ‘beat’. That is, if we understand correctly, Finn ‘black’ $<$ Baltic ‘bruised’. This is unlikely, however. That seems to us too marginal an acceptance of ‘black’ and too inaccurate as a colour designation to be the origin of a general term for ‘black’. There is another Finnish adjective *museva* ‘dark, blackish’ (SES), but it lacks the passive suffix *-ta-* and thus cannot be interpreted as ‘beaten’. Finn *mustelma* ‘bruise’ can easily derive from *musta* ‘black’ (through *muste* ‘ink’) but this does not mean that the reverse change is likely. To account for the difference between Balt *š* and Finn *s*, Liukkonen needs a sporadic free variation between *s*

7.3.4. Conclusions

To sum up, the evidence we have found for Baltic **s* > Fennic **š* boils down to very little:

- Not a single example after **ũ* (except for *tuhat* itself).
- Nouns in final **-īs* (< **-ijas*) > Fennic **-iš* (7.3.3.1). This means that even after **i*, Fennic **š* need not point to a RUKI reflex in Baltic.
- Two words in internal **-is-* (*laiha*, *vihvilä*, 7.3.3.2). This may be due to either “heightening” of **s* (ibid.), or to mere palatalization... or to chance.
- A few other words (*hanhi*, *pahla*, *lahto* and the distorted *ahingas*, 7.3.3.2). Various factors can be imagined, but not confirmed. The cluster *st* is normally allowed in Finnish.

Some of these loans may be later than *tuhat*.

Counter-evidence is even more scarce; possible examples are *kausta*, *laiska*, *musta* but none of them is certain (7.3.3.3).

The contradiction between *tuhat* and *kausta*, *musta* (if one or both of the latter were really from Baltic) would be annoying but need not worry us because various factors may have been at play.

All in all, our position is that Finnish *tuhat* is best explained as a loan from Baltic, that the Baltic source form was most probably **tūsantī* (or **tūstantī*) with **s*, and that there are a few parallels for a sporadic rendering of Baltic **s* as **š* in Fennic.

This view is far from resting on a firm ground, but we still advocate it, because the main alternatives would be either an early split of (East) Baltic, with full RUKI affecting not only Slavic but also the language that was the source of the Baltic loans in Finnish, or a non-IE origin of the word, two scenarios we find even less likely (7.3.1). See also 7.2.6.

7.4. **š* vs. *st*

We used to think that the lack of the medial *t* in Fennic **tušante* vs. Lith *tūkstantis* was due to a dissimilation (cf. 3.3.1, 5.1.3).

and *š*, which he sees in Lith *blūšti* ~ *blūsti* and **plauš-* ~ **plaus-*, referring to papers by Čekman and Karaliūnas respectively. We have not checked these papers but only the entries in Smoczyński 2018. According to Smoczyński, *blūsti* ‘blink (etc.)’ derives from the onomatopoeia *blūst* (also *blūst*, *blāst*). This makes the variation *s* ~ *š* in this word irrelevant. A relation between *plāusti* ‘rinse’ and *plaušas*, *plaušai* ‘fiber (of flax, hemp), bast’ is not obvious; the latter seems closer to *plūšti* ‘fray, be torn apart, wear out’. (In his paper written in Lithuanian, Liukkonen does not translate the Lithuanian words end roots, which did not help us understand his hypotheses.)

However, since we have been led to posit an early East Baltic stage **tūsantī* (5.5.1.1), this form can be the direct source of the Fennic word, provided the loan can be dated early enough, which is indeed the case until evidence to the contrary. Note that the short *u* in Finnish points to an early loan, too (7.2), although we don't know exactly how early is "early" in either case.

7.5. The nasal

For a long time, our word was posited with **-mt-* in Baltic, and accordingly in Fennic, at the time of the borrowing.

At first, this was mostly for the etymological reason we have already seen (the derivation from 'hundred', 4.5).

Thomsen 1893: 232 f. cites the Baltic forms including OPr *tusimtons* and the Fenno-Volgaic forms including Mari, but he does not propose any reconstruction of an earlier stage (except for the Fennic nominative singular [***]*tuhansi*).

Gauthiot 1906 insists (against Hirt's objections) on the etymology **tūs-k(o)mt-* and posits **tūšamt-* as the original Lithuanian form; he also cites the Mordvin and Mari forms, but it's not clear if he wants to derive the Mari *-m* from the old **-mt-*.

Būga 1908: 138 posits a pre-Lithuanian **tušamtīs*, again for etymological reasons (**tūs-kmt-*), but he does not cite the Fenno-Volgaic forms, except for Finnish "*tuhansi*" (apparently taken from Thomsen, see fn. 73). Būga presumably did not know Gauthiot's 1906 article, nor his Paris talk of 1904 (published later that year), when he wrote his big 1908 book, mainly based on his notes written until the autumn of 1905 (preface, p. v f.).

Nieminen 1957: 190, too, posits a primitive Baltic form **tūšamtis* (**tūšimtis* in West Baltic), whence a Proto-Fennic **tušamte*, but he does not cite forms of other Fenno-Volgaic languages.

Vaillant 1958: 647, on the other hand, derives Finnish "*tuhansi*" from an earlier [Fennic] "**tūšanti-*" with **n*, but he does not give a reconstruction of earlier Baltic forms.

SKES V 1374 (1975) writes Proto-Baltic **tūšamt-*, Proto-Fennic **tušamte-*, and posits an "[*m*]" in the pre-forms of Mordvin and Mari.

Lühr 1993: 124, 1998: 897 gives the Finnish pre-form as **tušamte-*, presumably on the authority of SKES (not Stang), although she posits the early Baltic form as **tūšantis* with *n* (897 f.; more clearly 1993: 124) and does not mention the Volgaic forms.

Liukkonen 1999: 90, *en passant*, posits a dialectal Baltic **tūšamtē* (see also 7.6) > Fennic **tūšamte* (with later shortening of **ū*, 7.2.3), presumably on the

authority of *SKES*, as it doesn't cite the Volgaic forms nor any Indo-European etymology.

Gorbachov 2006: 4, 13 and Jakob 2024: 140, 144, also posit **-mt-* in both the Baltic source and the Fenno-Volgaic rendering, but only on account of the final *m* in Mari *tüžem*. Both authors have rightly abandoned the old etymology of 'thousand', so they have to attribute the Baltic purported labial nasal to a secondary influence of 'hundred', even though 'hundred' had **i* (**šimta-*) where 'thousand' had **a* at the time of the borrowing (**tūsan̄tī* or similar). In our view, this influence was impossible, because of the different vowel (and different sibilant), and the Baltic source must have had **-nt-*.

Van Pareren 2008: 127⁽¹⁸⁾, quoting Stang 1966: 100, is aware that *mt* in Old Prussian cannot be old.

7.6. The final

Finnish *tuhat*, gen. *tuhannen*, plural *tuhannet*, gen. *tuhansien*, ordinal *tuhannes*⁷³; Estonian *tuhat*, gen. *tuhanden*, ordinal *tuhandes*; see *SKES* V 1374 for further Fennic forms.

The Baltic source of the Finnish loan was formerly mostly posited as **tūšamtis* (7.5), an *i*-stem like Old Lithuanian *tūkstantis*.

However, it is more likely that the Old Lithuanian form goes back to **tūstantī* (5.4.5:3, 5.5.2.1), which implies a still earlier **tūsan̄tī* at the time of the borrowing (5.5, 7.5).

Liukkonen 1999: 19 posits a dialectal Baltic **tūšamtē-* as source of the Finnic loan, but this is hardly possible because the modern dialectal *tūkstantē* is likely to be a *late* reshaping of the **jā*-stem, whereas the lack of a medial **t* in the Fennic loan points to an *early* stage (**tūsan̄tī*, 5.5.1) for the time of the borrowing.

In Finnish, apart from the nominative singular, the alternation *e/i/Ø* (and *te/de/si/t*) in the inflection of *tuhat* matches exactly that of inherited words like *joki* 'lake', *nimi* 'name' and more specifically *käsi* 'hand', *vesi* 'water' < Uralic **joke*, **nime*, **käte*, **wete*⁷⁴. In any case, since there was no (reconstructible) contrast

⁷³ We're not sure what Thomsen 1890-1893: 76 f., 233 means by *tuhansi* (without asterisk or hyphen, but between brackets), as opposed to the "usual" ("sædv.") nominative *tuhat* and stem *tuhante-*. This *tuhansi* is then cited by Būga 1908: 138, Vaillant 1958, 647, Fraenkel 1136, and others, as if it were the form (or one form) of the nominative singular, but as far as we have been able to enquire it's not the case and *tuhansi-* is only one of the stems of the plural.

⁷⁴ We conventionally write the final vowel as *-e* but there is currently no consensus on whether it was originally **e*, **i* ~ **ɪ* or **ə* (Pystynen 2017: 7).

between **i* and **e* for the stem-final vowel in Fennic⁷⁵, there is no reason to prefer **tūšamtē* or **tūšantē* over **tūšantī* as the Baltic source of Fennic **tušante*.

What's unclear to us is the apparently irregular loss of final **e* or later **i* even before the Fennic change **ti > *ci (> si)*.

Finally, the nasal was regularly lost before a syllable-final stop (cf. Posti 1953: 56-59).

7.7. Mordvin, Mari

As we have briefly noted (7.1.0), our word is also found in the Mordvin languages and in Mari.

7.7.1. Forms

Sources: (a) Wiedemann 1865: 161⁷⁶; Thomsen 99, 233, *SKES* V (1975): 1374; (b) Paasonen 1990 [1889-1901]⁷⁷, s. v.; Jakob 2024: 144.

Mordvin: Moksha *t'ožəń*, *-äń*, *-än*; Erzya (a) *t'oža*, *t'ožov*, (b) *t'ožań*, *t'užäń*⁷⁸, Kuldym village *t'ožəm*⁷⁹. – Erzya also uses *tyštsa* (thus Wiedemann), *tyštša*, *tysjača* (thus Gauthiot 1906: 1), from Russian тысяча, тыща⁸⁰.

Mari: West *tə-žem*, East *tüže-m*.

7.7.2. Comments

A number of remarks come to mind⁸¹:

(a) The Mordvin and Mari forms are remarkably similar despite the languages not being closely related. Perhaps the word has been borrowed from one of them into the other.

(b) Mordvin *t'*- is irregular. Inherited **u*, **ū* normally become Md *o*, *u* respectively (Bereczki 1988: 320 f.), but the preceding dental is unaffected (see

⁷⁵ The contrast between nominatives in *-i* and *-e* in Modern Finnish is only due to the presence, in the latter, of a former final **x < *š*, cf. 7.3.3.1.

⁷⁶ Probably the ultimate source for the following. (Thanks to A. Jakob, 2025 b, for the references to Weidemann and Paasonen.)

⁷⁷ The dictionary published in 1990 is based on materials collected during four field trips between 1889 and 1901, see map in Paasonen 1990 : LXIII. (Thanks to Jakob 2025 b for the reference and PDF.)

⁷⁸ *a* in the former and *u* in the latter are irregular. – It appears from the examples given in the entry that *t'ožań* is from a Moksha-born Erzya speaker. – (Both observations by Jakob 2025 b.)

⁷⁹ In the inflected (?) form *t'ožməń* (perhaps for *t'ožmäń* according to the dictionary).

⁸⁰ Russian words can be freely used in Erzya sentences, Raun 1988: 97. Apparently this was already the case in Weidemann's time.

⁸¹ See also van Pareren 2008: 126-129 for another discussion of the Mordvin loan and a comprehensive bibliography.

Rédei I 534-538, II 800 f. for **tu-*). Mari *ü* is irregular as well, for expected **u* (cf. Bereczki 1988: 338). *SKES* V 1374 asks whether the front vocalism in Mari is due to *šü·δə* ‘hundred’ and the palatal initial in Mordvin to *śado* ‘hundred’.

(c) On the one hand, the final *m* in Mari *tüžem* (and even Kuldym Moksha *t’ožəm*) should be old, because final *m* easily becomes *n* (or presumably *ń* in a palatalizing context?) and not vice-versa. On the other hand, however, the origin of this *-m* cannot be sought in Baltic, which had **nt*, not **mt* (4.5, 6.3, 7.5). Even if the Baltic source somehow had **mt*, its expected Fennic rendering may well have been **nt*. And even if early Fennic or Fenno-Volgaic had **mt*, there is little doubt that its Mari reflex would have been **nt* > *n*⁸². We see no other choice than to consider the Mari labial as an innovation. Perhaps there was a propagation of the labial feature of the vowel of the first syllable?

(d) Erzya *t’oža* might have its final from Russian *týščā*, which is also used in Erzya (7.7.1). However, loss of final **ń* is also found in some other words (van Pareren 2008: 125, Jakob 2025 b) and perhaps does not even need an explanation.

(e) Erzya *-v* in *t’ožov* ‘‘would seem to imply **-ŋ*’’ (Jakob 2025 b).

(f) Suhonen 1988: 614 observes that it is not necessary to assume that Baltic loanwords in Mordvin and Mari have reached these languages through Fennic mediation; and that Mordvin has some Baltic loans that are not found in Fennic. In the word for ‘thousand’, however, the three branches agree closely in that they all reflect a protoform in **-ũša-* as against East Baltic **-ũsta-* < **-ũsa-* < **-ũsi-* (4.2, 5.5.1). This is best explained if the word was borrowed from early Fennic (**tušante* or **-i*). Presumably Mordvin borrowed it from Fennic (indirectly⁸³) and Mari from Mordvin (cf. a).

7.8. Conclusion (Fenno-Volgaic)

We tentatively maintain that early Fennic **tũšante* (final vowel uncertain) was borrowed from early Baltic **tũsantĩ* (or **tũstantĩ*). The main alternatives would be an extinct Far-Eastern or Far-North-Eastern Baltic dialect with **-ũš-* (to account for **š*) or even a non-Indo-European origin of this word (to account for the short **u*), both of which we deem even more unlikely. There are a few possible parallels for this shortening of **ũ* (7.2) and this rendering of **s* as **š* (7.3) in Baltic loans. Even so, these substitutions are not well proven and not well explained.

There are reasons to believe that the Mordvin word was borrowed from Fennic and the Mari word from Mordvin (7.7). This is not formally proven, and

⁸² However, we have not checked the reflex of **pt* in Mari (and Mordvin).

⁸³ Before the expansion of Russian, several cultures have left archaeological traces between the present-day territories of Baltic or Fennic and Mordvin languages, part of them Fenno-Ugric, part of them not but perhaps Baltic, see van Pareren 2008: 74 f.

even if it were that would not entail a similar conclusion for other loans listed by Suhonen 1988: 614 or van Pareren 2008: 84-133 .

The question of a palaeo-Baltic substrate fully remains in spite of our removal of the word for ‘thousand’⁸⁴. To cite only one study, most of the other examples discussed or mentioned by Jakob 2024 are of a different sort in that they have no Slavic counterpart and no discernible IE etymology.

8. Etymology

8.1. Root

8.1.1. Forms

Our word is currently mostly seen as a derivative **tuh_x-s-(o)nt-* of the IE root **teuh_x-* ‘to swell, to grow thick, big, strong’, cf. Ved *TAV¹* ‘be strong, have power’, *tuvi-* ‘very’, *tavás-* ‘strong’ < **teuh_x-és-*; on this root see Mayrhofer I: 638 f., *LIV*² 639 f.)⁸⁵; for this etymology (even before the laryngeal theory) see Bopp 1853: 46 and e. g. Lühr 1993: 125 (chosen as an example of “Semantifizierung von Zahlwörtern”), Gorbachov 2006: 10; further references in Pijnenburg 1989: 100 under (2) and Lühr 1998: 894 f.

Others have seen it as a compound of the same root with a word for ‘hundred’ (see Pijnenburg l. c. under (1) and Lühr l. c. 892 f.), which amounts to the same semantically speaking, but does not account for the Germanic and Slavic *-s-* (5.1.1).

A relation of ‘thousand’ to this root ‘big, strong’ is only denied, to our knowledge, by those few scholars who hold for a non-Indo-European origin, be it for formal reasons only (namely, the apparent impossibility of tracing the Germanic, Slavic, Baltic and Fennic forms to a single preform if this is sought for in Indo-European: thus Jakob 2024: 144 f.), or, additionally, because the semantic connection is allegedly unfounded (“schwebt in Wirklichkeit in der Luft”, Stang 1972: 59, quoting Pedersen 1943: 185).

Now, a non-IE origin does not solve the formal difficulties (it gives no explanation for the divergent East Baltic form, at least, 5.3).

⁸⁴ And ‘lax’, if from PIE **lok₁-os-/*lk₁-s-* with Lith *lašiša* < **-šas-* < **-šas-*, and perhaps by metaphor Latin *locusta* ‘grasshopper’.

⁸⁵ Laryngeal **h₁* in our opinion (Francis’ Law) if Gk grec τῆρός ‘cheese’ is related; **h₂* according to Peters 1980: 290²⁴³, who relates Gk σάος ‘safe and sound’ < **twawos* < **tuh₂-ewo-*. None of these two etymologies is semantically satisfying, however.

8.1.2. Semantics

As for the semantic connection (*‘big, strong’ > ‘thousand’), it can be supported by a number of parallels.

(a) ‘big’ > an unspecified big number:

– Lith *tūlas* (sg. or pl.) ‘several, many’, cf. OPr *tūlan* ‘much, a lot’, from the same root PIE **tuh_x*-⁸⁶;

– French *force* ‘strength’ can be used in the sense of ‘many, a goodly number of’ (e. g. *avec force remerciements* ‘with profuse thanks’; this is a remnant of medieval syntax and semantics in the sense of ‘une quantité de’, cf. *FEW* III: 728 f.).

– Gk ἄλλος ‘in abundance, in throngs, enough’, from the same PIE root as English *swell* according to Imberciadori 2023.

(b) an unspecified big number > a definite big number:

– Gk μῦριοι ‘countless’ > μῦριοι ‘ten thousand’ (Chantraine s. v.).

(c) ‘big’ > a definite big number

– English *grand* subst. ‘a thousand of some unit of currency’, from *grand* adj. ‘large, great’;

– French *grosse* subst. ‘twelve dozens’ (thus not as big a number as ‘thousand’, however), feminine of *gros* adj. ‘groß, dick ; big, fat’.

(d) ‘strength, size’ > ‘number’:

Armenian *t’iw* ‘number’, plausibly from **tēwes-o-* *‘relative to strength’ from an assumed sigmatic neuter **tewes-* < **teuH-es-* ‘strength’ (cf. Olsen 1999: 23).

8.2. *-s-suffix

a) The sigmatic stem is perhaps the zero grade of the sigmatic neuter deverbative **teuh_x-os* ‘bigness’, or perhaps an enlarged root **tuh_xs-* that is seen in some Germanic and other words (Pokorny 1084).

A substantive **téuh_x-os* n., Skt. **távas-* n. is not directly attested.

There are (with -s-):

In the Ṛgveda (Mayrhofer I: 638 f.):

– *tavás-* adj. ‘strong’,

– *tavás-* m. subst. ‘strength’, with suffixal accent (but “hardly substantive” according to Wackernagel-Debrunner 1954 §123cA, *mettre la page, et 884*),

⁸⁶ A connection with Slavic *tyl^o* ‘nape of the neck’ is uncertain, cf. on the contrary Gk ἀγχίην ‘id.’ < **ang^hw-én-* < **ang^h-ú-* ‘narrow’.

– *távasvant-* ‘strong’, as if from **távas-* ‘strength’ (adjectives in *-vant-* reflect the accent of the basic noun, with few exceptions that are not similar to *távas-vant-*, Wackernagel-Debrunner 1954: 884),

– *táviṣ-* ‘strength’ (**téuh_xs-ih₂*) and its own derivatives,

– *tuviṣ-*, *túviṣ-* (in derivatives, a contamination of **taviṣ-* and **tuvi-*).

In Armenian: *t’iw* ‘number’ (8.1.2 d) makes a neuter **téuh_x-os* plausible but a post-PIE creation is not excluded.

If a noun or adjective in **tuh_xs-* is derived from a noun **téuh_x-os*, it must go back to a time when sigmatic neuters still had an ablauting paradigm, of which there only remain two traces as far as I know, **h₂eus-os*, gen. **h₂us-és-s*, dual **h₂us-s-ih₁* ‘ear(s)’ (cf. Irslinger 2008: 339-342) and **h₃éh₁-os*, **h₃h₁-és-s* v. sim. ‘mouth’ (cf. Irslinger 2008: 387-390)⁸⁷.

b) Alternatively, an enlarged root **t(e)uh_x-s-* is perhaps to be recognized in a number of derivatives, like ON *pústa* ‘heap’, Old Irish *tuaimm* ‘hill’ (Pokorny 1084).

8.3. **-nt*-suffix

8.3.1. Parallels

Since only a feminine in **-ntī* can be safely reconstructed, while the *a* or **o* is probably or possibly not old in the Old Norse and Old High German variants *tusand*, *dūsant* (2.1), in regional Slavic **tysotji* (3.2) and in East Baltic **tūstantis* (5.4.5:4, 5.5.1.1), mere linguistic comparison does not tell us whether the full grade of the underlying masculine had been **tuHs-ont-*, **tuHs-ent-* or even **teuHs-ont-*.

The suffix of this **tuHs-nt-* ‘big, strong’ (to cite its zero grade) is found in a few other adjective (of related meaning), but their formations are different⁸⁸:

a) **b^hrǵh-ónt-* ‘high’ (Ved *brhánt-* ‘high, big, strong, etc.’; see Mayrhofer II: 232 for its cognates; note Arm *barjunk* ‘pl. ‘high’ < **b^hrǵh-ónt-es*, Stempel 1983: 92), synonym of **b^hrǵh-ú-* and others (see Mayrhofer l. c.), is merely the present participle of the v e r b **b^herǵh-* ‘to rise’ (*LIV*² 78 f., Schirmer–Kümmel), just as

⁸⁷ The ‘ear’ word is not attested in Tocharian and Anatolian, as if it were a creation of “Core Indo-European” (unless it). On the other hand, Anatolian has already levelled the alternation (in *nepiš*, *nepišaš* ‘sky’, like Skt. *nábhaḥ*, *nábhasaḥ*, Gk. *véφος*, *véφεος* ‘cloud’, and partly in *aiš*, *išaš* < **h₃eh₁-es*, **h₃h₁-es-os*). The leveling presumably took place independently in Anatolian and the rest of Indo-European. A verbal root for these three nouns is unknown, unlike the productive type **génh₁os* and the possible **téuh₂-os* discussed here.

⁸⁸ The following derives from this concise remark by Neri 2009: 8: “(...) ein sekundäres Adjektiv **tuh₂-s-ont-* ‘groß, angeschwollen’ (zu **teuh₂-s-* ‘Schwellung’) (...) (zu den sekundären *-ont-* Bildungen vgl. z. B. uridg. **m(e)ǵ-éh₂-ont-* → ai. *mahánt-*)”.

**h₁d-ónt-* ‘tooth’ is the (non-Anatolian) present participle of **h₁ed-* ‘to bite (repeatedly or lastingly), to eat’⁸⁹.

b) **m̥g-eh₂-ént-* (v. sim.)/**m̥g-h₂-nt-* ‘big’ is an enlargement of the adjective **meḡ-éh₂-* (or **-óh₂-*), neuter **méḡ-h₂* (Ved *mahām* acc., *máhi*, see Mayrhofer II: 337-339, Wodtko 2008: 468, 470-473), seen in Vedic *mahánt-/mahat-* ‘big’ and its Avestan cognate, Greek ἄγᾱν ‘too much’, Latin *ingēns* ‘huge’ (Mayrhofer l. c., Meier-Brügger 1979, Wodtko 2008: 469, 474 f.)⁹⁰. – For Mayrhofer l. c., Ilr **maj^h-aH-ant-* ‘big’ is analogical of **b^hj^h-ánt-* ‘high’, so that we would not have to care about the morphological function of the suffix. However, because of ἄγᾱν, *ingēns*, this analogy would have to go back to the proto-language, which requires **meḡ-eh₂-ónt-* or **m̥g-eh₂-ónt-* with **o:* this ancienty would make the full grade **-eh₂-* (or **-oh₂-*) even stranger and this vocalism makes the second α of ἄγᾱν perhaps more difficult to explain. See Wodtko 2008 l. c. for alternative views on this suffix.

c) Formally more archaic and semantically more remote: **ḡerh₂-ont-/ḡrh₂-nt-* ‘old’, an adjective (not a participle) from the verb **ḡerh₂-* ‘alt machen, let grow old’ (*LIV*² 165 s. Kümmel)⁹¹.

8.3.2. Conservative analysis

E. g. present participle **tūs(o)nt-* < **tuh₂s-(o)nt-*, Lühr 1998: 898, 901; denominal adjective **tūs(o)nt-* < **tuH-s-(o)nt-*, Neri 2009: 8; or only its feminine

⁸⁹ **h₁ed-* is morphologically a “present root”, i. e. durative; ‘bite repeatedly’: cf. Lat. *mandūcāre* ‘masticate’ > French *manger*; ‘bite lastingly’: cf. **h₁ed-won-/h₁d-un-* ‘pain’ (Schindler, Lamberterie

⁹⁰ See Meier-Brügger l. c. 155, 157 for the Greek long ā. The rebuttal of Osthoff’s Law in final position by Peters 1980: 309 f. is mistaken. – The connection of Lat *ingēns* with Ved *mahánt-*, Gk ἄγᾱν was discovered in the late 19th c, but rejected in the early 20th until it was revived by Meier-Brügger 1979, now followed by Wodtko l. c., cf. de Vaan s. v., and others. – This etymology of Lat. *ingent-* (< **engant-*) involves a semantic difficulty because its meaning is not just the neutral ‘big’ but an expressive, emphatic ‘huge, immense, enormous’. The obstacle can be taken care of, however, with the assumption of a semantic attraction by negative compounds like *immēnsus* ‘immeasurable, boundless, huge, enormous, infinite’ (mostly “with some degree of hyperbole”, *OLD*), further away *immānis* ‘savage, frightful; of enormous size, tremendous, very great’, *īnfīnītus* ‘indefinite, unrestricted; in hyperboles: endlessly long, infinite in quantity or amount, innumerable, immense, extreme’; with a different prefix *ēnormis*. Perhaps *ingēns* was reinterpreted as a negative compound itself, with or without a sense being given to the second element.

⁹¹ NOT ‘aufreiben, attrite’, a concoction of Grassmann’s faithfully handed down for generations. Mycenaean and Laconian Greek **geronsiā* was the council of elders, not of decrepit oldsters; Non-Balkan European **ḡrh₂-no-* ‘grain, corn’, is not from *‘zerrieben’ (that would mean ‘flour’!), but from ‘which has ripened’ (de Vaan 2008 s. v. *grānum*).

**túsntī* < **túHs-nt-ih₂*, Gorbachov 2006: 10, 13; further views discussed by Pijnenburg 1989: 103 f, Lühr 1993: 121.

8.3.2.1. Notes

Un participe présent est supposé par Lühr 1993, 124, Casaretto 2004, 441 ; mais il faudrait un présent athématique au vu de l'alternance **-ont-/*-nt-*, or il n'existe guère de présents en **R(Ø)-se-*. Un participe aoriste conviendrait mal pour le sens (aspect ponctuel) et peut-être pour le vocalisme radical.

Neri 2009, 8 suppose un adjectif dénominatif du type de v. i. *mahānt-/mahat-* < i.-ir. **majaHant-/*majHat-*, mais ce dernier a un vocalisme différent et n'est pas entièrement clair lui-même. Pour Wackernagel-Debrunner II/2, 166 f., Mayrhofer II, 337 f. (1994), il s'agit d'une innovation indo-iranienne, élargissement de **majáH-* < **meg-oh₂-* d'après **b^hrǵ^h-ánt-* 'haut'. Pour Meier-Brügger 1979, de Vaan 2008, 303, Wodtko 2008, 469, 474 f., la formation est déjà indo-européenne et source du latin *ingēns* 'énorme, immense'. Ce dernier rapprochement serait sémantiquement improbable (passage d'un terme neutre 'grand' à un terme expressif/emphatique) si l'on ne pouvait supposer une attraction des composés négatifs *immēnsus* 'sans limites, immense, démesuré, infini' et plus lointainement *immānis* 'monstrueux, gigantesque, immense', *īnfīnītus* 'sans fin, illimité, infini'. Le degré plein serait peut-être **mǵ-éh₂-ont-* ; l'initiale i.-ir. *ma-* serait une réfection de **a-* < **m-* d'après le simple **majaH-* (v. i. *mahā-*) < **meg-oh₂-*.

D'autres adjectifs en **-ont-* sont **b^hrǵ^h-ónt-/-nt-* 'haut', doublet de **b^hrǵ^h-ú-*, qui a bien le degré radical zéro mais est déverbatif, et **gérh₂-ont-/*ǵrh₂-nt-* 'vieux' (Mayrhofer II, 576, av. litt.), déverbatif également.

À cause du **s*, **tuHs-(o)nt-* peut difficilement être déverbatif et devrait donc être dérivé de **teuH-és-* 'fort', skr. *távas-*, ou de **téuH-es-* 'force', cf. avest. *paiti-tauuah-* '...' et le dérivé arm. *t'iw* 'nombre' s'il vient de **tēwes-o-* ← **téwes-* < **téuH-es-* (cf. Olsen ...déjà citée). Dans les deux cas, cependant, le degré radical zéro suppose un accent suffixal **tuHs-ónt-*.

Cette analyse, dans la ligne de celle de Neri (ci-dessus), reste possible si l'on suppose que le terme de quantité **tús(o)nt-* 'mille' (ou plutôt : 'grande quantité', ...), un substantif, est dérivé de l'adjectif **tūs(ó)nt-* par la "loi des appellatifs", à la manière dont gr. *λεῦκος*, nom d'un poisson, *λεύκη* 'peuplier blanc' sont dérivés de *λευκός*, *λευκή* 'blanc, blanche' (cf. Vendryes 1904, 152 f.).

8.3.3. Risky analysis

If the Northern IE word for 'thousand' is derived from **teuh_x-es-* 'big size, strength', the expected suffix would rather be **-went-* (or **-wont-*) 'having,

endowed with’, yielding **tuh_{xs}-wnt-ih₂* > **tūswntī*. (Similar formation in RV *távas-vant-* ‘strong’, although this is an independent, later creation rather than a reshaping of **tuh_{xs}-went-*.)

8.3.3.1. Phonetics

The loss of **w* would be rather unproblematic in Balto-Slavic, which has other instances of loss of **w* after a sibilant (OCS *sestra* ‘sister’, Lith *sesuõ* ‘id.’, *šėšuras* ‘father-in-law’, *sāpnas* ‘sleep’) or even after **k* (*večer̃s*, *vākaras* ‘evening’ vs. Arm *gišer* ‘night’ < **wekwero-*, Viredaz 2017: 6). (The loss is independent in B and S: *šėšuras* ≠ *svek̃r̃s*.) In our word this could be a dissimilatory loss due to the preceding **ū*.

As for Germanic, it is known that **w* is preserved before **u* in initial position (many examples: Kroonen 2013: 597-601) as well as after consonant when supported by the analogy of full-grade forms (as in past participles like English *swollen*, *swum*, German *gezwungen* ‘forced’ < OHG *gidwungan*). However, there seem to be a few examples of loss⁹²:

**ga-sunda-* ‘sound, healthy’, probably akin with Gmc **swinþa-* ‘strong’ (Kluge–Seebold s. vv. *gesund*, *geschwind*)⁹³.

**sunda-*, ON *sund* n. ‘a swimming’, OE *sund* ‘power of swimming’ (hence also Hence also ON *synda* ‘swim’, *syndr* ‘able to swim’, OE *syndig* perh. ‘skilled in swimming’), vs. Goth *swumfsl* ‘pool’ (a more recent creation, or *w* preserved because link still felt with ‘swim’. (Wiktionary: “possibly with restored w”).)

Thus, if these two etymologies are correct, **tūswntī* > **pūsundī* would be regular.

(Should they not, a dissimilation from the preceding **ū* might be pressed to duty.)

8.3.3.2. Morphology

Zero grade before the suffix **-went-* is possible not only if the basic noun had zero grade but also if it had an alternation between full grade and zero grade (cf. Wackernagel-Debrunner 1954: 885 for Sanskrit), e. g. Ved. *kṣumánt-*

⁹² WGmc **þurh* ‘through’ is not an example of **w*-loss, cf. Goth *þairh*, OE *þerh*, *þurh* ‘id.’ < **þerxwe* (Kroonen 2013: 538). It may well be related to **þwerxa-* ‘cross’ (adj.), but the latter is probably from **þerxwa-* (Kroonen 2013: 556).

⁹³ Seebold himself (l. c.) derives the latter from a stem **sġh-went-* ‘strong’. If correct, this could be added to our §8.3.1. However, there does not seem to have been a root noun with the meaning ‘strength’. The root **seġh-* rather meant ‘get a firm hold of’, hence either ‘to defeat, conquer’, or ‘to possess, to have’. A meaning ‘strong, strength’ only developed in a few derivatives, see Wodtko 2008: 600 f.

‘abounding in food, strong (etc.)’, Avestan *fšūmant-* ‘keeping cattle’ (Mayrhofer I: 433), from **péku/*pku-* ‘bétail’.

Such an alternation in the sigmatic noun **téuh_x-os/*tuh_x-s-* would point to a very archaic stage (cf. 8.2 a end), before the leveling of the productive type to generalized **R(e)-es-* (except for nom.-acc. sg. **R(e)-os* in non-Anatolian).

The reconstruction **tuHs-wnt-ih₂* does not seem to yield cognates in other IE languages any more than **tuHs-nt-ih₂*⁹⁴.

8.3.3.3. Dialectology

**-went-* (or **-wont-*) is only attested in Anatolian, Indo-Iranian, Greek, while **tūs(w)ntī* is found in none of them. Could it be that **-wEnt-* was a dialectal feature in “Indo-Hittite” and still was in “Core Indo-European”?

In a conservative, Stammbaum-centered view, one would think that if a feature is old enough to be found in Anatolian, it is also old enough to have been proto-“Core Indo-European” and thus common to all “Core Indo-European”.

But perhaps PIE already had a few dialectal nuances when it experimented its major split between Anatolian and the rest, and perhaps that split crossed some pre-existing variations. Note that this would imply that the Anatolian split resulted from more than one migration.

Thus speculation derives from the observation that Luwian is Satem-like in its reflex of **k̑*, and even Aryan-like in its vocalism, with Čop’s law as a kind of pendant of Brugmann’s law (because both point to PIE **e* being shorter than PIE **o*). These could be just two instances of parallel development, but the fact that in both cases Luwian sides with Indo-Iranian while Hittite sides with European gives to think. However, it would take more than two instances to even have the beginnings of a proof.

In short: in a conservative view of dialectology, the discrepancy between the areas of **tūsntī* and that of **-went-* is no obstacle to our etymology **tūswn̥ti*, because an earlier pan-European extension of **-went-* (and even **tūswn̥ti* in its original meaning of ‘huge number’ can be taken for granted. In the freer view we advocate in the case of Luwian and Aryan above, on the contrary, said discrepancy speaks against our own analysis of the ‘thousand’ word. This dialectological question will be left open at least here.

⁹⁴ It goes without saying that if this new reconstruction is correct, **tūsntī* in previous paragraphs must be changed to **tūswn̥ti* in all instances where it refers to early stages (prior to the presumably independent loss of **w* in Germanic and Balto-Slavic), and **tuHsntih₂* to **tuHswntih₂* where appropriate (at least not in quotations nor in 8.3.2).

8.4. Accent

The accent of the feminine adjective must have been on the final **-ih₂*.

On the other hand, the accent of the substantive ‘thousand’ was on the first syllable as shown by Germanic. (In Balto-Slavic, the initial accent might be due to Hirt’s Law.)

This accent shift is a regular marker of the substantivization (Vendryes 1904: 152-154, 160 f., Dieu 264-271).

The case is not the same as in Gk μῶριοι (adj.) ‘countless’ > μῶριοι (adj.) ‘ten thousand’ (8.1.2 b).

Accordingly, the accent shift need not be linked to the new numeral meaning.

Nor do we think (as do Hamp 1968: 277 and probably many others) that the Northern type **tūsṇtī* took the place of earlier terms belonging to the **ǵ^heslo-* family.

It seems more plausible to us that the substantive **tuh_xs-ṇt-ih₂* was already Proto-Indo-European (8.2 a, end) but that it merely meant ‘a huge quantity’, and only took the numeral value ‘thousand’ in “North European” dialects, at the same period (or a little later) when “Southern” dialects attributed the meaning ‘thousand’ to some derivatives or compounds of **ǵ^hes-lo-* ‘hand(ful) (of seeds)’ (9.2 a).

9. ‘Mille’ dans les dialectes indo-européens

9.1. Formes

Le numéral ‘mille’ se présente ainsi dans les langues indo-européennes (hors anatolien⁹⁵ et tokharien⁹⁶) :

a) Termes apparemment dérivés de **ǵ^hes-lo-* (apparenté à **ǵ^hes-r-* ‘main’) et désignant à l’origine une poignée de grains de blé (Rix 1991); **ǵ^heslo-* aurait signifié ‘main’ (Rix) ou peut-être ‘poignée’ (à notre avis):

– Quasi **sm-ǵ^heslo-* > i.-ir. **sa-j^hásra-*, neutre thématique comme **cátá-* ‘cent’⁹⁷ ;

⁹⁵ Le mot ‘mille’ ne nous est pas connu: Eichner in Gvozdanović 1992:; wiktionary:

⁹⁶ Tocharian had its own term, A *wälts* B *yaltse* < **weltyo-* or perh. **welso-*, cf. sl. **velbjb*, **velikb* ‘grand’, Adams 2013, 523 av. litt. – A *tmām*, B *tmāne*, *t_umāne* ‘myriade’ est emprunté au turc ou directement au chinois, ibid. 318 s.

⁹⁷ Les deux mots ont été empruntés en finno-ougrien **cata* et **carsa* (Rédei 1988: 466 f., avec l’ancienne restitution **ś*. Le second ne reflète pas forcément un plus archaïque **j^hásra-* (qui contredirait l’analyse sémantique de Rix 1991: 230 f., mais serait compatible avec **ǵ^hes-lo-* ‘poignée’) : il pourrait aussi avoir perdu la première syllabe **sa-* par une forme de simplification ou de dissimilation.

– Quasi **ǵ^hesl-io-* > proto-grec **k^héhlloi*, adjectif comme *χ-κόσιοι* (dor. -κάτιοι) ‘x cents’ et *μύριοι* ‘dix mille’ ;

– P.-ê. quasi **smih₂ ǵ^hesl-ieh₂* > **smī ǵ^hesliā* f. → **-i* (neutre comme *centum*) > lat. *mīlle*, pl. *mīlia* (**smī-ǵ^hslī*, Sommer fdate, **smī-ǵ^heslī* > **mīhēli-*, Szemerényi 1954, **smī ǵ^hesliā*, Hamp 1968). Ce juxtaposé serait antérieur à l’élimination des laryngales, car **-ih₂* final devient *ia* dans les monosyllabes en latin (*tria*, *qui*; de même en germanique, cf. **prijō*, et peut-être en celtique).

Hamp 1968: 277 f. further speculates that the Greek and Latin types are substitutes for a pan-IE **sm-ǵ^heslo-* preserved in Indo-Iranian, but we hardly see more basis for this than the strive for monogenesis (2.1.6) rather than factual support.

b) Termes apparemment dérivés de **tuH-* ‘gros, fort’ : germanique et slave **tuHs-ŋt-ih₂* f. > **tūsŋtī* (2, 2.5), baltique **tuHs-ont-* > **tūsont-* m. ou n. (6).

£ Tocharian (n. 96): Meillet, Antoine (1911-1912). “Les noms de nombres en tokharien B.” *Mémoires de la Société de linguistique de Paris* 17: 281-294 : *velbjb* ou *velikb* aussi ‘1000’ ? > éventuellement mettre dans le Handout corrigé

c) Les autres branches ne peuvent pas être classées :

– Le terme hérité (s’il a existé) a été remplacé par des emprunts en celtique insulaire (p. ex. gall. *mil*, v. irl. *míle*), albanais (*mijë*), arménien (*hazar*).

– Le mot ‘mille’ n’est pas attesté en celtique continental, dans les langues italiques autre que le latin, en phrygien.

9.2. Comments

1) = 9.1

2) Given its etymology, the word **tūsŋtī* (or even **tūsŋtī*) must go back to Proto-Indo-European and thus be just as old, if not older, than the derivatives and compounds of **ǵ^hes-lo-*.

3) The precise dialectal extension of **tūs(w)ŋtī* is unknown. Although **tūsŋtī* is often cited as the paramount example of a Baltic-Slavic-Germanic isogloss, let’s note that we don’t know what word for ‘thousand’, if any, was used in Tocharian, Armenian, Albanian and Celtic before they resorted to loans, nor what words were used in Anatolian.

4) Still, **tūs(w)ŋtī* has a more “compact” dialect area than the “Southern” group of words, which points to an innovation, that is, to the *meaning* ‘thousand’ being a late, dialectal specialization of ‘huge number’.

10. Conclusions

10.1. Summary of conclusions

– Germanic and Slavic may reflect the single protoform **tūsŋtī* (this is not new: 2, 3).

– So may Baltic (Gorbachov, modified: 5.5, 6).

– East Baltic: perhaps **tūsintī* → **tūsantī* → **tūstantī* → **tūstantis* (~ **-jā*) > *tūkstantis* (5.5, 10.1).

– Old Prussian: either from “Elbing” High German **tūsant* (Smoczyński, modified), or from Early Baltic **tūsintī* above (6).

– Finnic **tušante* from East Baltic **tūsantī*. We’ve tried to argue that the shortening of *ū* and the retraction of **s* are possible in such a loan (7).

– Root: **teuH-* ‘thick, big, strong’ (this is not new). While the primary verb meant ‘to grow big’ (*LIV*²), the numeral cannot derive from **‘growing big’* and must have been built on a nominal derivative, such as **teuH-es-* ‘bigness’, at a time when sigmatic neuters still had ablaut.

– Suffix: rather than **-ent-* or **-ont-*, which do not yield a good morphological interpretation, perhaps the adjective was in fact **tuHs-went-* ‘endowed with bigness’, feminine **tuHs-wŋt-ih₂* > **tūswŋtī*, with the sporadic loss of postconsonantal **w* known in Balto-Slavic and a conditioned loss of **w* in Germanic (8.3).

– Initial accent (proved by Germanic) due to the substantivization of the adjective **tuHs-wŋt-ih₂* (8.4)

– If this etymology is correct, the noun **tuHs-ŋt-ih₂* or **tuHs-wŋt-ih₂* ‘huge number’ must already have been Proto-IE (9).

– The semantic specialization to ‘thousand’, on the other hand, may well have been post-PIE in both the “South” and the “North” (cf. 9).

– Note the unity of the “Northern” term **tūswŋtī* as opposed to the variety of the “Southern” terms **g^hesl-io-*, **sm̃-g^heslo-*, **smī-g^hesl-iā* (9).

– The precise dialectal distribution of the “Northern” and “Southern” terms is unknown, because the old terms (if any) have given way to loanwords in Celtic, Albanian, Armenian, Tocharian and the word for ‘thousand’ is unknown in Continental Celtic and in Anatolian languages (9).

10.2. Diachrony of *tūkstantis*

**teuh_x-* ‘to swell, grow thick, big, strong’ (*LIV*² p.)

> **teuh_x-os* n. ‘big size, strength’ (cf. Skt *távas-vat-* ‘strong etc.’; weak stem **tuh_x-s-* in Pre-PIE, § où je citerai ‘oreille’ 8.3.3.2; alternatively, stem **tuh_x-s-* of uncertain origin and function, 8.3.2)

> **tuh_xs-wént-* adj. ‘big, strong’ (vérif. accent)

> **tuh_xs-wnt-ih₂* (fem. of preceding; not sure what nouns it was used with)

> **túh_xs-wnt-ih₂* ‘huge quantity or number’ (substantivization of preceding, 8.4)

> “North PIE” **túswntī* ‘a thousand of’ (fem.) (vs. various “South PIE” derivatives of **ǵ^hes-lo-*, 9.1)

> Baltic **túswintī*, Slavic **túšwintī*, Germanic **túswuntī*

(**s* vs. **š*: 4.2; loss of **w* independently in B, S, and G, 8.3.3.1)

> a) Baltic **túsintī* (see below),

b) Slavic **túšintī* (> **túčintī* after **činta-* ‘hundred’, > PSI I **túsintjī* > PSI II **týsętji* > variant **týsętji*, 3, spec. 3.2.2),

c) Germanic **túšuntī* (later **pūsunþī* > **pūsundī*, 2.2.2)

> East Baltic **túšantī* (after the unmarked present participle suffix, 5.4.5 4, 5.5.1.1; West Baltic: *either* still **túsintī*, later **túsimtī* after **simtan* ‘hundred’, 6.5; *or* the OPr word is borrowed from German, *ibid.*)

> **tústantī* (perhaps by hypercorrection on the model of the inchoative present participle fem. *-stantī* if this was occasionally reduced to **-santī* in fast speech by dissimilation, 5.4.4, 5.5.1.2)

> *tústantis* (Bretkūnas, hapax; change of inflection, perhaps because **-ī* > *-i* was unfrequent except in adjectives, 5.4.5: 3 c; alternative change to *-ančia*, dial. *-antė*, or further change to masc. *-is*, gen. *-io*, *ibid.*)

> *tūkstantis* (sporadic *k*-insertion, 4.1.1; an additional influence of the root *tūk-* ‘grow fat’ is unlikely after all, 4.1.2).

11. Bibliography

- Adams, Douglas Q., 2013: *A Dictionary of Tocharian B* ². 2 vol. (¹1995) Amsterdam/New York.
- Adolphi, Henricus, 1685: *Erster Versuch einer kurtz-gefassten Anleitung zur Lettischen Sprache*.
- Aitzetmüller, Rudolf, 1978: *Altbulgarische Grammatik als Einführung in die slavische Sprachwissenschaft*. Freiburg i. B.: Weiher.
- ALEW: *Altlitauisches etymologisches Wörterbuch*, online version, <<https://alew.uni-frankfurt.de/>>
- Bereczki, Gábor, 1988, “Geschichte der wolgafinnischen Sprachen”, in: Sinor (ed.) 1988: 314-350.
- Bezenberger, Adalbert, 1877: *Beiträge zur Geschichte der litauischen Sprache auf Grund litauischer Texte des XVI. und XVII. Jahrhunderts*, Göttingen.
- Bopp, Franz, 1853: *Über die Sprache der alten Preussen in ihren verwandtschaftlichen Beziehungen*. Berlin
- Braune, Wilhelm, et alii, 2023: *Althochdeutsche Grammatik* 1¹⁷ (¹1886).
- Brugmann, Karl: *Grundriss der vergleichenden Grammatik der indogermanischen Sprachen*, Strassburg: Trübner.
1886: I¹, 1897: I²/1-2, 1906: II²/1, 1909: II²/2/1, 1911: II²/2/2.
- Brunner, Karl, 1942: *Altenglische Grammatik*. Halle S.: Niemeyer.
- Būga, Kazimieras, 1908: *Aistiški studijai (Aistische Studien)*. St. Petersburg.
- Bugge, Sophus, 1888, “Etymologische Studien über germanische Lautverschiebung”, *PBB* 13: 311-339.
- Casaretto, Antje. 2004. Nominale Wortbildung der gotischen Sprache. Die Derivation der Substantive. Heidelberg: Universitätsverlag C. Winter. (Indogermanische Bibliothek. Dritte Reihe: Untersuchungen).
- Chantraine, Pierre, 1933: *La formation des noms en grec ancien*, Paris: Klincksieck. Reprint 1979.
- Cleasby–Vigfusson 1874: *An Icelandic-English dictionary*, based on the ms. collections of the late Richard Cleasby. Enlarged and completed by Gudbrand Vigfusson. Oxford: Clarendon.
- Comrie, Bernard, 1992: “Balto-Slavonic”. In: *IEN*: 717-833.
- Derksen, Rick, 1996: *Metatony in Baltic*.
- , 2008: *Etymological dictionary of the Slavic inherited lexicon*, Leiden : Brill.
- , 2015: *Etymological dictionary of the Baltic inherited lexicon*, Leiden : Brill.
- Diels, Paul. 1963: *Altkirchenslavische Grammatik*. Heidelberg: Winter.

- Dieu, Éric, 2022: *Traité d'accentuation grecque*, Innsbruck: Institut für Sprachwissenschaft der Universität.
- Endzelīns, Jānis, 1923: *Lettische Grammatik*.
- , 1971: *Comparative Phonology and Morphology of the Baltic Languages*. Translated by W. R. Schmalstieg and B. Jēgers. La Haye/Paris. (Latvian original 1948, Lithuanian translation 1957.)
- FEW*: Walter von Wartburg et al., *Französisches Etymologisches Wörterbuch*, 1928–.
- Fraenkel, Ernst, 1953: “Hypernormalismen besonders im Slavischen”, *ZfslPh* 22: 88-122.
- , 1962-1965: *Litauisches etymologisches Wörterbuch*. 2 vol. Heidelberg : Winter and Göttingen : Vandenhoeck & Ruprecht.
- Garde, Paul, 1980: *Grammaire russe I: Phonologie – Morphologie*. Paris : Institut d’études slaves.
- Gauthiot, Robert, 1906: “Finnois: *tuhā*”. *JSFO* 23/14: 1-4.
- Gorbachov, Yaroslav, 2006: “North Indo-European” **tuh_ssntih₂* ‘1000’. A new proposal. [Conference handout.] AATSEEL Annual Meeting, Philadelphia, 29 December 2006.
- , 2014: “The origin of the Baltic inchoative in *-sta-*. An overlooked Proto-Baltic sound law”, *IF* 119: 21-53.
- von Greyerz, Otto & Bietenhard, Ruth, 1984: *Berndeutsches Wörterbuch*³. Bern: Francke (¹1976).
- Grimm, Jakob, 1853: *Geschichte der deutschen Sprache* I². Leipzig: Hirzel.
- Hamp, Eric P., 1968: “Latin *mille*”, *Glotta* 46: 274-278
- , 1973: “North European ‘1000’”: In: Cl. Corum, T. Cedric Smith-Stark, A. Weiser (eds.), *Papers from the IX Regional Meeting of the Chicago Linguistic Society*, April 13-15, 1973. Chicago: Chicago Linguistic Society, 172-178.
- van Helten, Willem, 1900: “Zu den malbergischen Glossen und den salfränkischen Formeln und Lehnwörtern in der Lex Salica”, *PBB* (Halle) 25: 225-542.
- 1906: “Zum germanischen Zahlwort”. *IF* 18, 84-126.
- Hirt, Herman, 1896: “Akzentstudien. 1. Germ. got. *pūsundi*”. *IF* 6: 344-349.
- IEN*, 1992: Jadranka Gvozdanović, *Indo-European Numerals*. Berlin - New York: Mouton de Gruyter.
- Imberciadori, G. (2023). On the etymology of Greek ἄλις ‘in abundance, in throngs; enough’, en Cuadernos de Filología Clásica. Estudios griegos e indoeuropeos 33, 27-37.

Irslinger 2008: entries in *NIL*.

Jakob, Anthony, 2024: “The palaeo-Baltic substrate: a methodological exploration”. In: *E litoribus Balticis etymologiae*, 123–162. Uralica Helsingiensia 15. Helsinki 2024.

—, 2025: personal communications (a: Aug 11; b: Sept 22).

Kallio, Petri, 2015: “The stratigraphy of the Germanic loanwords in Finnic”, in: John Ole Askedal & Hans Frede Nielsen (eds.), *Early Germanic Languages in Contact*: 23–38. NOWELE Supplement Series 27. Amsterdam - Philadelphia, PA.

Kalniņš. Aigars, 2016: *The Baltic sta-present and the Outcome of PIE *sḱ in Baltic* Diss. Copenhagen (Betreuer: Thomas Olander).

Kluge 1901, “Vorgeschichte der germanischen Dialekte”, in Hermann Paul (ed.), *Grundriss der germanischen Philologie* I², Strasbourg: Trübner: 320–496.

Kluge–Seebold, 1995: Seebold, Elmar, *Kluge: Etymologisches Wörterbuch der deutschen Sprache*. Bearbeitet von Elmar Seebold. 23., erweiterte Auflage. Berlin-New York : de Gruyter.

Koivulehto, Jorma. 1984: “Itämerensuomalais-germaaniset kosketukset” [Ostseefinnisch-germanische Kontakte]. In Åström, Sven-Erik (ed.), *Suomen väestön esihistorialliset juuret* [Die prähistorischen Wurzeln der finnischen Bevölkerung]: 191– 204. Helsinki: Societas Scientiarum Fennica.

Kroonen, Guus, 2013: *Etymological Dictionary of Proto-Germanic*. Leiden/Boston : Brill.

Kurschat, Friedrich, 1876: *Grammatik der littauischen Sprache*, Halle: Waisenhaus.

Leumann, Manu, 1942: “Idg. *sḱ* im Altindischen und im Litauischen. 3. Die litauischen *st*-Praesentien”. *IF* 58: 113–130. (Reprinted in his *Kleine Schriften*, 1959.)

Liukkonen, Kari, 1994: “Baltų dalyviai suomių kalboje” (Baltic participles in Finnish), *Baltistica* 4: 90–91.

*LIV*², 2001: *Lexikon der indogermanischen Verben*² (ed. H. Rix), Wiesbaden.

Lühr, Rosemarie, 1993: “Zur Semantifizierung von Zahlwörtern: Das Wort ‚Tausend‘ – eine germanisch-baltoslavische Isoglosse?” *Linguistica* 33: *Bojan Čop septuagenario in honorem oblata*, 33: 117–136.

<https://www.researchgate.net/publication/307718845_Zur_Semantifizierung_von_Zahlwortern_das_Wort_’tausend’_-_eine_germanisch-baltoslavische_Isoglosse>

[—], 1998: “*dūsunt*”. In : *Etymologisches Wörterbuch des Althochdeutschen* II (Hgg. Lloyd, Lühr, Springer). Vandenhoeck & Ruprecht : 890–902. <<https://ewa.saw-leipzig.de/articles/d%C3%BBsunt/>>

- eMED: Middle English Dictionary*, online database for the period 1100–1500, <https://quod.lib.umich.edu/m/middle-english-dictionary/dictionary>.
- Martinet, André, 1955: *Économie des changements phonétiques*, Berne : Francke. Réimpr.
- Mausser, Otto, 1933: *Mittelhochdeutsche Grammatik auf vergleichender Grundlage*- München: Hueber. Reprint 1972.
- Mayrhofer, Manfred, 1986-2001: *Etymologisches Wörterbuch des Altindiarischen*. 3 vol. Heidelberg : Winter.
- MED A Middle-English Dictionary, containing words used by English writers from the twelfth to the fifteenth century*, by Francis Henry Stratmann. A New Edition, revised, rearranged, and enlarged by Henry Bradley. Oxford: Clarendon, 1891. Reprints: OUP.
- Meier-Brügger, Michael, 1979: “Griechisch (μηδὲν ἄγαν”, *MSS* 38: 155-163.
- Meillet, Antoine, 1923: *Caractères généraux des langues germaniques*². Paris: Hachette. (¹1917).
- , 1924: *Le slave commun*¹.
- Meillet, Antoine & Vaillant, André, 1934: *Le slave commun*². Reprint 1965.
- , 1980: *Grammaire de la langue serbo-croate*² (¹1924).
- Morgenstierne 1926: *Report on a linguistic mission to Afghanistan*, Instituttet for sammenlignende kulturforskning, Oslo.
- Morgenstierne 1975: *Irano-Dardica*, Wiesbaden: Reichert: chapter “Die Stellung der Kafirsprachen”, 325-377.
- Neri, Sergio, 2009, review of Don Ringe, *From Proto-Indo-European to Proto-Germanic*, in *Kratylos* 54 (2009), 1-25.
- Nieminen, Eino, 1957: “Über einige Eigenschaften der baltischen Sprache, die sich in den ältesten baltischen Lehnwörtern der ostseefinnischen Sprachen abspiegelt”, *Proceedings of the Finnish academy of science and letters* 1956: 185-206. Helsinki: Academy.
- NIL*, 2008: Dagmar S. Wodtke, Britta Irslinger, Carolin Schneider, *Nomina im Indogermanischen Lexikon*, Heidelberg: Winter.
- Noreen, Adolf, 1904: *Altschwedische Grammatik*. Halle: Niemeyer.
- Olsen, Birgit Anette. 1999. *The Noun in Biblical Armenian, origin and word-formation*. Berlin-New York: Mouton De Gruyter.
- Paasonen 1990: *Mordwinisches Wörterbuch*.
- van Pareren, Remco, 2008: “Die direkten baltischen Lehnwörter im Mordwinischen”, *Finnisch-ugrische Mitteilungen* 30/31: 69-147.

- Pedersen, Holger, 1943: “Et baltoslavisk problem”. In: R. Brøndal, V. Brøndal, Chr. Møller, H. Olsen (eds.), *In memoriam Kr. Sandfeld*. Udgivet paa 70-aarsdagen for Hans Fødsel. København: Gyldendalske Boghandel Nordisk Forlag, 184-194.
- Peters, Martin, 1980: *Untersuchungen zur Vertretung der indogermanischen Laryngale im Griechischen*. Wien: ÖAW.
- Petit, Daniel, 2000: “Quelques observations sur les substantifs de genre neutre en vieux prussien”. *Baltistica* 35: 29-43.
- , 2010: *Untersuchungen zu den baltischen Sprachen*. Leiden, Boston: Brill.
- , 2022: “Traduction et oralité dans l’*Enchiridion* vieux-prussien (1561)”, *BSL* 117: 217-247.
- Pijnenburg, W. J. J., 1989: “Eine germanisch-baltoslawische Isoglosse”, *HS* 102: 99-106.
- Plöger, Angela, 1982. “Über die Entstehung des finnischen Stammtyps *CVC(Ca/ä)*”. *Finnisch-Ugrische Forschungen* 44: 66–98.
- Pokorny, Julius: *Indogermanisches Etymologisches Wörterbuch* (1959–1969).
- Pystynen, Juho, 2022: “New lines of evidence on the secondariness of Finnic *ii, *uu”. *HELSLANG Summer Conference · 2022-05-24*.
https://www.academia.edu/79912362/New_lines_of_evidence_on_the_secondariness_of_Finnic_ii_uu
- Raun, Alo, 1988: “The Mordvin language”, in: Sinor (ed.), 1988: 96- 110.
- Rédei, Károly (dir.), 1988–1991: *Uralisches etymologisches Wörterbuch*. 3 vol. Wiesbaden: Harrassowitz.
- Rix, Helmut, 1991: “Urindogermanisch **ǵheslo-* in den südindogermanischen Ausdrücken für ‘1000’”. In: *Studia etymologica Indoeuropaea memoriae A.J. van Windekens (1915-1989) dicata* (ed. L. Isebaert), Louvain: 225-231.
- Ross, Alan & Berns, Jan, 1992: “Germanic”. In: *IEN*: 555-715.
- Ruggiers, Paul G., 1979: *The Canterbury Tales. Geoffrey Chaucer. A Facsimile and Transcription of the Hengwrt Manuscript, with Variants from the Ellesmere Manuscript*. University of Oklahoma Press and Dawson, Folkestone, UK.
- Seebold: see Kluge–Seebold
- Senn, Alfred, 1966: *Handbuch der litauischen Sprache*. Heidelberg: Winter.
- SES: *Suomen etymologinen sanakirja* [Finnish etymological dictionary]. Kotimaisten kielten keskus [National Languages Center].
<https://kaino.kotus.fi/ses/>. Accessed Sept-Oct 2025

- Sinor, Denis (ed.), 1988: *The Uralic languages: description, history and foreign influences*. Leiden etc.: Brill, 1988.
- SKES: *Suomen kielen etymologinen sanakirja* [Etymological dictionary of the Finnish language. 7 vol. 1975-1983.
- Smoczyński, Wojciech, 2018: *Lithuanian Etymological Dictionary*, Berlin etc.: Lang.
- Sommer, Ferdinand, 1899:£ “titre” *IF* 10: 216 f.
- Stang, Christian S, 1966: *Vergleichende Grammatik der Baltischen Sprachen*. Oslo: Universitetsforlaget.
- , 1972. *Lexikalische Sonderübereinstimmungen zwischen dem Slavischen, Baltischen und Germanischen*. Oslo: Universitetsforlaget.
- Stempel, Reinhard, 1983: *Die infiniten Verbalformen des Armenischen*. Frankfurt a. M.-Bern: Lang.
- Streitberg, Wilhelm, 1896: *Urgermanische Grammatik*. Heidelberg: Winter.
- , 1920: *Gotisches Elementarbuch*⁵⁻⁶. Heidelberg: Winter.
- Suhonen, Seppo, 1988: “Die baltischen Lehnwörter der finnisch-ugrischen Sprachen”. In: Sinor (ed.) 1988: 596-615.
- Szemerényi, Oswald, 1954: “The Indo-European cluster *sl* in Latin”. *Archivum Linguisticum* 6: 38-41. Reprinted in his *Scripta Minora* II, 1987: 711-725.
- , 1990: *Einführung in die vergleichende Sprachwissenschaft*⁴. Darmstadt (¹1970).
- , 1996: *Introduction to Indo-European Linguistics*. Transl. of 1990 by Irene and David Morgan Jones. Oxford: OUP.
- Thomsen 1890-1893
- Trautmann, Reinhold, 1910: *Die altpreußischen Sprachdenkmäler*. Reprint Göttingen: Vandenhoeck & Ruprecht, 1970.
- Ulmann, Carl Christian, 1872 *Lettisches Wörterbuch*. Erster Theil: Lettisch-deutsches Wörterbuch. Riga: Brutzer.
- de Vaan, Michiel, 2008: *Etymological dictionary of Latin and the other Italic Languages*, Leiden: Brill.
- Vaillant André, 1950: *Grammaire comparée des langues slaves*. I. Phonétique. Lyon–Paris, IAC.
- , 1958: *Grammaire comparée des langues slaves*. II. Morphologie. Première partie : Flexion nominale. Lyon–Paris, IAC.
- Vasmer, Max & Trubačov, Oleg: *Ètimologičeskij slovar’ russkogo jazyka*. Moscow, 1964-1978.

- Vendryes, Joseph, 1904: *Traité d'accentuation grecque*, Paris, Klincksieck.
Réimpr. Villers-sur-Marne, Phénix, 2003.
- Viredaz, Rémy, 1997: “‘Six’ en indo-européen”. *IF* 102, 112-150.
- , 2017: “Notes d’étymologie arménienne III”, *REArm* 37 (2016-217): 1-20.
- , 2020: “Notes d’étymologie slave ». In: *Etymologus. Festschrift for Václav Blažek*, ed. Harald Bichlmeier, Ondřej Šefčík, Roman Sukač. Hamburg : Baar: 403-422.
- Vondrák, Wenzel (Václav): *Vergleichende slavische Grammatik* ¹1906, ²1924.
- Wackernagel, Jakob & Debrunner, Albert, 1954: *Altindische Grammatik* II/2. Nominalsuffixe. Göttingen : Vandenhoeck & Ruprecht.
- Walde, Alois & Pokorny, Julius. *Vergleichendes Wörterbuch der idg. Sprachen*. I, 1930, II, 1927. Berlin.
- Weber, Albert & Bächtold, Jacques M., 1983: *Zürichdeutsches Wörterbuch*³. Zürich: Rohr.
- Wiedemann 1865: Grammatik der Ersä-Mordwinischen Sprache
- Wodtko 2008: entries in *NIL*.
- Yrjö-Koskinen, Eino Sakari, 1900: *Dictionnaire finnois-français*. Helsinki : Impr. Suomalaisen Kirjallisuuden Seura
- Zinkevičius, Zigmas, 1980-1981: *Lietuvių kalbos istorinė gramatika*. 2 vol. Vilnius: Mokslas.